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Comprehensive Notes on Press and Cutting Tools Design

MEP 406

BY

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Introduction

Press working may be defined as a chipless manufacturing process by which various components are made from sheet metal. This process is also termed as cold stamping. The machine used for press working is called a press. The main features of a press are: A frame which supports a ram or a slide and a bed, a source of mechanism for operating the ram in line with and normal to the bed. The ram is equipped with suitable punch/punches and a die block is attached to the bed. A stamping is produced by the downward stroke of the ram when the punch moves towards and into the die block. The punch and die block assembly is generally termed as a “die set” or simply as the “die”. Press working operations are usually done at room temperature.

Press Operations

The sheet metal operations done on a press may be grouped into two categories, cutting operations and forming operations. In cutting

operations, the work piece is stressed beyond its ultimate strength. The stresses caused in the metal by the applied forces will be shearing stresses. In forming operations, the stresses are below the ultimate strength of the metal. In this operation, there is no cutting of the metal but only the contour of the work piece is changed to get the desired product. The cutting operations include: blanking, punching, notching, perforating, trimming, shaving, slitting and lancing etc. The forming operations include: bending, drawing, redrawing and squeezing. The stresses induced in the metal during bending and drawing operations are tensile and compressive and during the squeezing operation these are compressive.

Below we give the definitions of the various press operations:

- a. **Blanking:** Blanking is the operation of cutting a flat shape from sheet metal. The article punched out is called the 'blank' and is the required product of the operation. The hole and metal left behind is discarded as waste. It is usually the first step of series of operations, Fig. 2.1 (a).

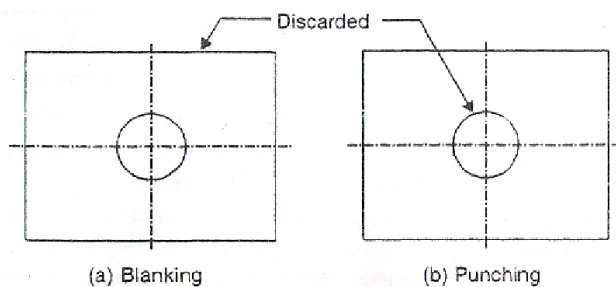


Fig. 2.1. Blanking and Punching.

Plain blanking is explained in Fig. 2.2

- b. **Punching (Piercing):** It is a cutting operation by which various holes are made in sheet metal. Punching is similar to blanking except that in

punching; the hole is the desired product, the material punched out to form the hole being waste Fig. 2.1 (b).

- c. **Notching:** This is cutting operation by which metal pieces are cut from the edge of a sheet, strip or blank.

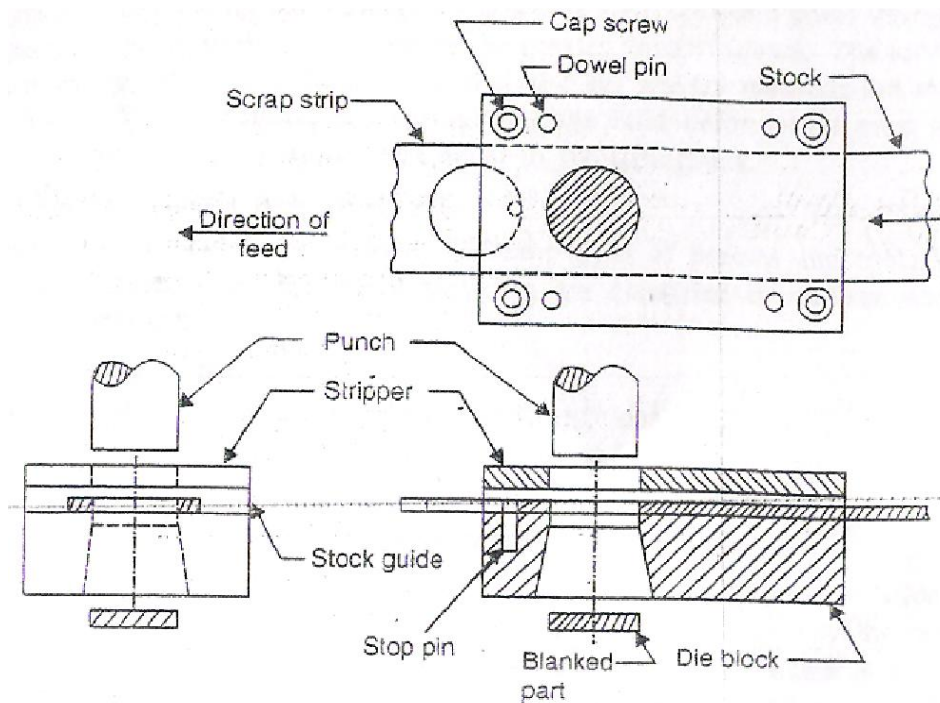


Fig. 2.2. Plain Blanking.

- d. **Perforating:** This is a process by which multiple holes which are very small and close together are cut in flat work material.
- e. **Trimming:** This operation consists of cutting unwanted excess material from the periphery of a previously formed component.
- f. **Shaving:** The edges of a blanked part are generally rough, uneven and unsquare. Accurate dimensions of the part are obtained by removing a thin strip of metal along the edges. The operation is termed as shaving.
- g. **Slitting:** It refers to the operation of making incomplete holes in a work piece.

- h. **Lancing:** This is a cutting in which a hole is partially cut and then one side is bent down to form a sort of tab or louver. Since no metal is actually removed, there will be no scrap.
- i. **Nibbling:** The nibbling operation which is used for only small quantities of components is designed for cutting out flat parts from sheet metal. The flat parts range from simple to complex contours. This operation is generally substituted for blanking. The part is usually moved and guided by hand as the continuously operating punch cuts away at the edge of the desired contour.
- j. **Bending:** In this operation, the material in the form of flat sheet or strip is uniformly strained around a linear axis which lies in the neutral plane and perpendicular to the lengthwise direction of the sheet or metal.
- k. **Drawing:** This is a process of forming a flat work piece into a hollow shape by means of punch which causes the blank to flow into a die cavity.
- l. **Squeezing:** Under this operation, the metal is caused to flow to all portions of a die cavity under the action of compressive forces.

Some of these operations are explained in Fig. 2.3

As written above, the work piece obtained after one or more press operations is called a metal stamping. A metal stamping can be defined as one which is usually made of, but not limited to, sheet or strip metal that has been cut, coined, pierced, bent, formed or drawn in one or more operations between matching dies under pressure. The size of a metal

stamping, the thickness, the type of material and shape vary widely. Metal stampings range in size from tiny instrument part to freight car ends.

The materials amenable to cold stamping are low-carbon and plastic alloyed steels, non-ferrous metals including copper and its alloys (brass, bronze), aluminium, zinc, titanium and other materials. The starting material for press working is sheet metal in the form of bands or strips from tenth of a mm to about 6 or 8 mm thick. During the stamping process, the wall thickness of the parts remains almost constant and differs only slightly from the thickness of the initial sheet metal. The material must be free from metal. The material must be free from scale before sheet metal working. Scale is removed by pickling and other methods.

The use of metal stampings is not confined to any single industry. On the contrary, a large percentage of manufactured products of all types employ at least one or more metal stamping.

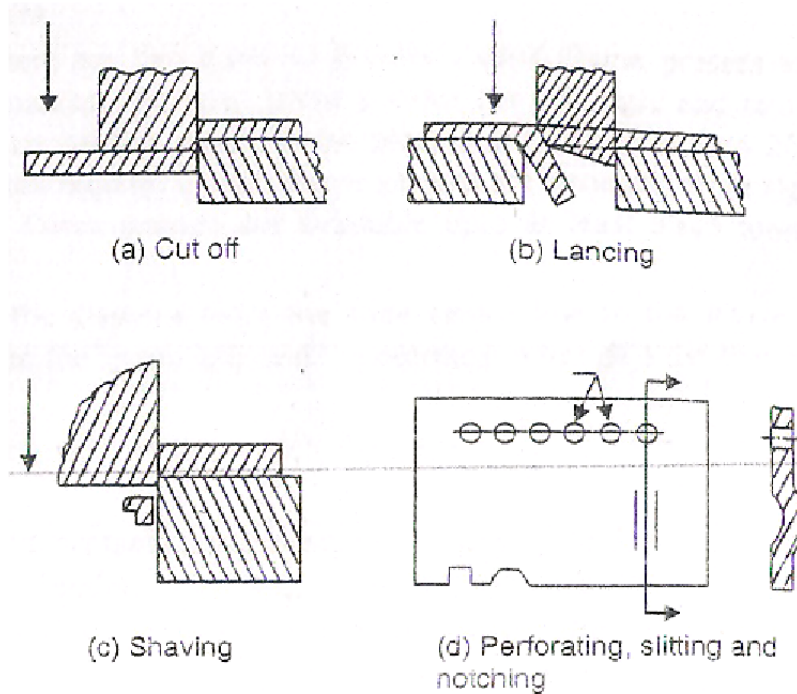


Fig. 2.3. Some Press Operations.

Advantages of Metal Stamping

Sheet stamping is an advanced technique of mechanical working. It has got the following advantages.

1. Small weight of fabricated parts.
2. High productivity of labour (up to 30,000 to 40,000 parts per 8 hour shift).
3. High efficiency of technique as regards the fabrication of items of diversified shapes both simple and complex, such as washers, bushings, retainers (cages) of ball bearing, tanks and car bodies.
4. The parts made by cold sheet metal stamping are distinguished for their size accuracy (narrow tolerances and high surface finish). In many cases, they require no subsequent machining and are delivered to the assembly shop.

5. High-volume, low-cost production.
6. Predictable strength characteristics.
7. Uniformity of parts.
8. Low cost material.
9. Less labour-consuming.

All the above advantages have made sheet stamping very attractive to a host of industries, particularly to automotive, aircraft, electrical engineering and others. The variety of products is immense, from aircraft skins and automobile bodies to application shells. Other products include: construction girders, truck frames, furniture legs, supertankers, bath tubes, beer cans, wheel rims, fan blades, watch gears and so on (list is very long).

Elastic Recovery or spring back

We know that in metal working processes, the total deformation imparted to a work piece will be the sum of elastic deformation and plastic deformation. We also know that elastic deformation is recoverable whereas plastic deformation is permanent. So, at the end of a metal working operation, when the pressure on the metal is released, there is an elastic recovery by the material and the total deformation will get reduced a little. This phenomenon is called as “spring back”. This phenomenon is more importance in cold working operations, especially in forming operations such as bending etc. spring back depends upon the yield point strength of a metal. The higher the yield point strength of a metal, the greater the spring back. The amount of spring back account for it. To

compensate for spring back, the cold deformation must always be carried beyond the desired limit by an amount equal to the spring back.

PRESS-WORKING EQUIPMENT

Press-working equipment includes different types of presses and tools, known as dies or 'press-tools'. Presses used for metal stamping are classified in various ways. They may be classified according to:

- i. Sources of power
- ii. Method of actuation of ram/rams (Slide/Slides)
- iii. Number of slides
- iv. Type of frame
- v. The type of work for which the press has been designed.

The two main sources of power for applying force to the ram or slide of a press are: mechanical and hydraulic. In mechanical presses, the linear movement of the ram is obtained with the help of a flywheel driven system. The heavy flywheel absorbs energy continuously from an electric motor and delivers it to the ram and hence to the work piece intermittently. Hydraulic presses are a large piston and cylinder arrangement coupled to a hydraulic pump. The piston and the press ram are one unit. The ram is actuated by oil pressure on the piston in the cylinder. Water and water-oil emulsion can also act as working mediums.

Mechanical press has the following advantages over hydraulic presses:

- (1) Run faster (2) lower maintenance cost (3) lower capital cost

Advantages of hydraulic presses are:

1. Most versatile and easier to operate
2. Tonnage capacity adjustable from zero to maximum
3. Constant pressure can be maintained throughout the stroke
4. Force and speed can be adjusted throughout the stroke, if need be
5. More powerful than mechanical presses
6. Safe as it will stop to a pressure setting
7. Stroke can be limits of hydraulic cylinder travel.
8. The press at any position of the ram stroke.

The main disadvantage of the hydraulic press is that it is slower than a mechanical press.

The flywheel of the mechanical press drives the main shaft. The rotary motion of the main shaft is converted into the linear motion of the ram or slide. The most common drive is crankshaft which is used for longer strokes. The eccentric drive can be used for smaller strokes. The other drives include: Cam, toggles, rack and pinion, screws and knuckles which can be used to actuate the ram. Cam drive is similar to eccentric drive except that at the bottom of the stroke, dwell is obtained. Rack and pinion drive is used where a very long stroke is required. In this drive, the movement of the slide is uniform and much slower as compared to crank drive.

Most press working makes use of mechanically driven and less frequently, hydraulic presses. According to the number of slides, there are three types of presses: single, double, or triple action press. A single action press has only one ram which acts against a fixed bed. A double action press has two slides (one inner and the other outer) moving in the same direction against the fixed bed of the press. Such presses are used in deep drawing operations. A triple action press has two slides moving in the same direction against the fixed bed as in double action press and the third or lower slide moves upward through the fixed bed. Such presses are used for large work such as motor car body panels.

According to the type of frame, there are two types of presses: open frame presses and closed frame presses. Due to their construction, the open frame presses are less rigid and strong and so are useful mainly for operations on smaller work. These presses are available up to 200 tone capacity with strokes of 90 to 120 per minute. Closed frame presses are stronger, more rigid and balanced than open frame presses. These presses are available up to at least 3000 tones capacity.

The “Throat Depth” of a press is the distance from the slide centre line to the frame. It determines the maximum distance between the shank axis and the rearmost point of a die that can be mounted on the press.

Small presses often have inclinable press frames which facilitate removal of the stamped part by gravity.

Depending upon the type of work, the presses can be designed for doing the following types of work: Punching, Blanking, Drawing, Bending, Forming, Coining, Embossing etc.

According to the method of strip feeding the presses can be designed either for manual or automatic feed.

According to the method of removing finished parts, there are three types of presses.

1. Presses with dies with a hole through which the part drops.
2. The ones in which the component is ejected upwards and then removed by a positive knockout.
3. Those in which the finished parts are removed by compressed air or manually.

Rating of a Press

A press is rated in Tones of force that can be applied to the slide without undue strain and without affecting the structural strength of the press.

The tonnage of a mechanical press is determined by the size of the bearing for the crankshaft or the eccentric. It is given approximately by the relation.

Tonnage capacity = Shear strength of the crankshaft material x the area of the crankshaft bearings

The tonnage capacity of a mechanical press is always given when the slide is near the bottom of its stroke, because it will be maximum at that point, or with the crank turned through an angle of not more than 30° from the bottom zero position.

The tonnage capacity of a hydraulic press is given as:

Rated capacity = Piston area x oil pressure in the cylinder.

As noted earlier, the capacity of a hydraulic press can be varied by changing oil pressure. In double-action crank presses, the tonnage of the inner slide determines the maximum drawing pressure, while the maximum blank holding pressure depends upon the tonnage of the outer side. To keep the strains and the deflections of the press structure small, it is a usual practice to choose a press rated 50 to 100% higher than the force required for an operation.

In the case of single, double, and triple action hydraulic presses, the rams may all be driven from a central hydraulic accumulator fed by pump, or have individual drives from one or more pumps.

Requirements of Press Tool Design

The press tool design should suit the type of production it will be used for, that is, small batch, large batch, or mass production. The press tools should meet the following requirements:

1. The dimensional accuracy and surface finish of stampings should conform to the drawings and specification.
2. The working parts of the press tool (die and punch) must be adequately strong, durable in operation and easily replaceable when worn out.
3. The die should ensure the required hourly output, easy maintenance, safe operation and reliable fastenings in the press.
4. The die should be designed in such ways that as far as possible standard components are used for its manufacture. As few special parts as possible should be used in its design.
5. The scrap in the stamping operation must be kept at a minimum by suitably designing the strip layout. The percent utilization of the material is given as:

$$\text{Percentage material utilization} = \frac{\text{Total area of blank cut}}{\text{Area of uncut strip}} \times 100$$

Normally, the value of this factor is 70 to 80%.

Press Tool Components

The press tool components may be divided into the following types, see Fig. 2.4

- a) Working components, which participate in the shaping of parts: dies, punches and their sections.

- b) Structural components, which serve the joining, the working components to one another and to the press: Upper shoe (Punch holder), lower shoe (die holder) and shanks.
- c) Guiding components, which ensure accurate alignment of the upper shoe with the die shoe in operation: guide posts and bushings. Guide posts also facilitate tryout of the press tool in a press.
- d) Feeding components, which feed the stock strip or blanks to the stamping station.
- e) Locating and locking components, which provide for an accurate positioning of the stock or blank in the die and fix it in place while the operation is performed?
- f) Stripping components, which strip and remove the blanks and scrap from the working components after the operation is over: strippers push off pins, knockouts.
- g) Fastening components, which join and hold together all parts and units of the press tool: punch plates, die-blocks and cases, and all fasteners?

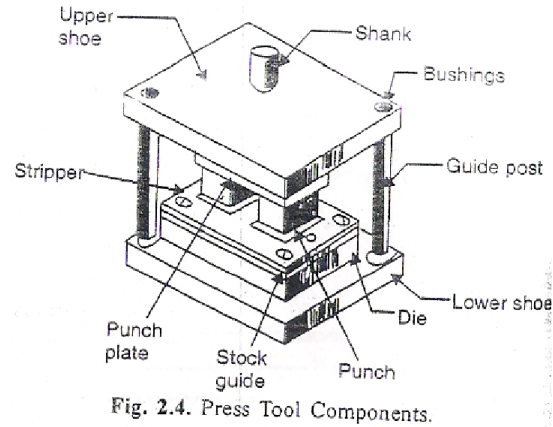


Fig. 2.4 is a schematic representation of a press tool comprising all principal components present in any typical press tool. The shank secures the press tool to the press ram. It fits in the clamping hole in the lower end of the press ram.

Building Up of the Press Tool

In spite of the great variety of press tools, most of them have certain components similar in purpose and design. To facilitate design, manufacture, maintenance and repair of press tools, similar units and components have been standardized.

The most important standardized component of a press tool is “Die-Set”, Fig. 2.5, which consists of an upper shoe (punch holder), lower shoe (die-holder) and guide posts. Punch (or punches) is mounted on the upper shoe and the die components are mounted on the lower shoe (die shoe).

The available space for mounting punch and die components is called the “die-area”. The die area on the lower shoe should be at least 6.35 mm larger all around the die block. Ears are provided on the flange of the die shoe to fasten it to the bolster plate or the bed of the press, with the help of 7-bolts.

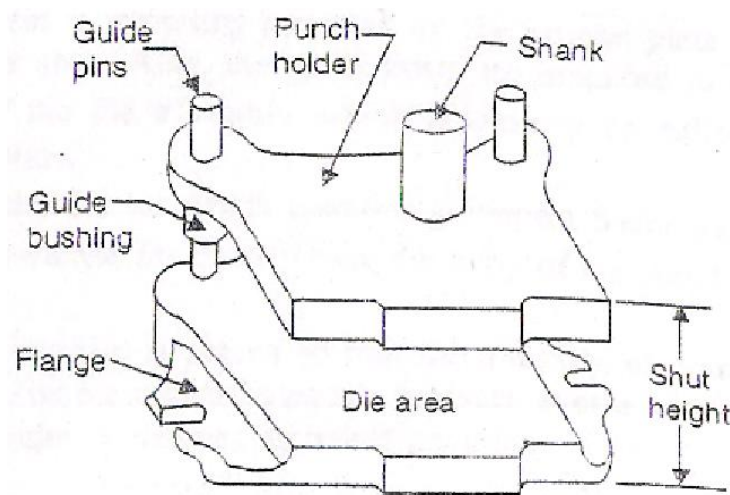


Fig. 2.5. A Commercial Die Set

In smaller die set the upper shoes are made with a shank. On longer die sets (for cutting press tools) the standard shouldered shanks are press fitted into a specially bored hole of the upper shoe, Fig. 2.6 (a). In drawing and bending press tools, threaded shanks are used Fig. 2.6 (b). There are other arrangements also for fitting the shanks into the upper shoes of

larger die sets. If the tool shank does not fit in the press slide hole, adapter shanks or split adapter bushings may be employed.

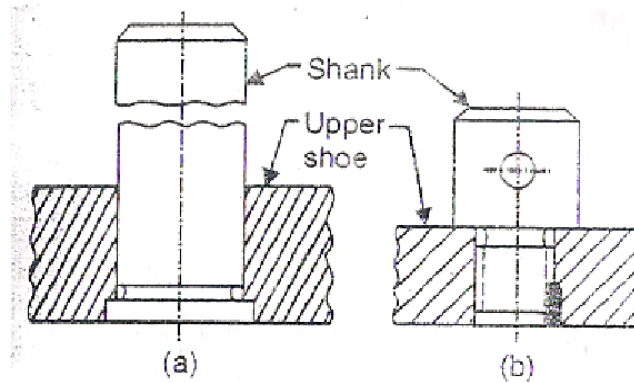


Fig. 2.6. Press Tool Shanks.

Arrangement of Guide Posts

The various arrangements of guide posts are given below:

- a) **Back-post design:** In this design, the die-set has two guide posts positioned in the rear, Fig. 2.7 (also see Fig. 2.5).

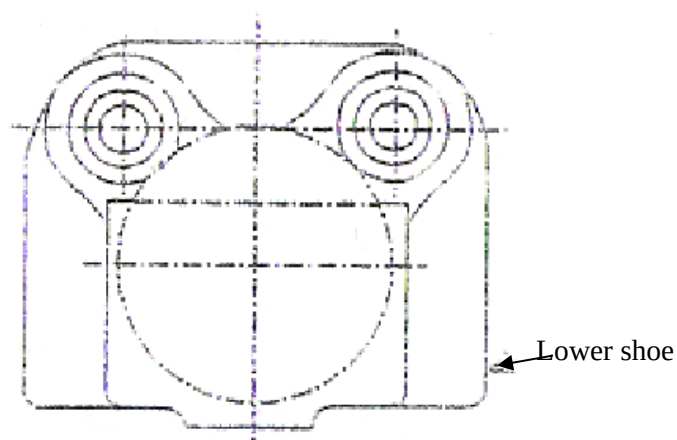


Fig. 2.7. Back Post Design

- b) **Centre-post or Axial-post design:** In this design, the two guide posts are positioned along the transverse centre line of the die-set, Fig. 2.8. This design is suitable when the load is too heavy and the strip feeding is from the front. This design prevents end feeding.

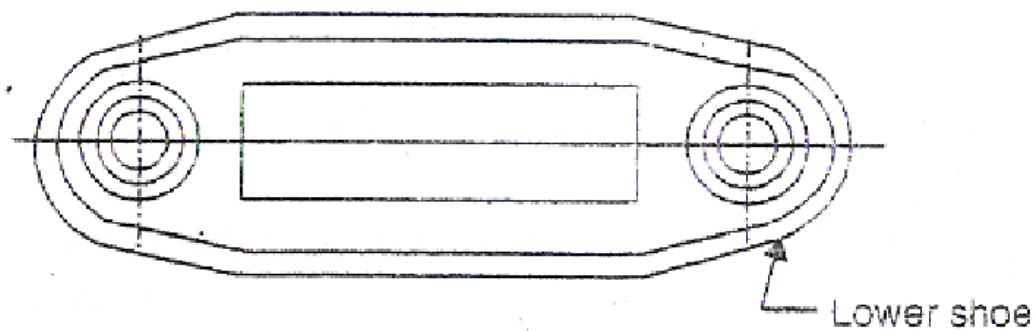


Fig. 2.8. Axial-post Design.

- c) **Diagonal-post design:** This design, Fig. 2.9 is used when the load is very heavy and end feeding is required. The guide posts are along a diagonal line. Generally, the left guide post is in front, but if feed is to be from left to right, the right guide post is in front.

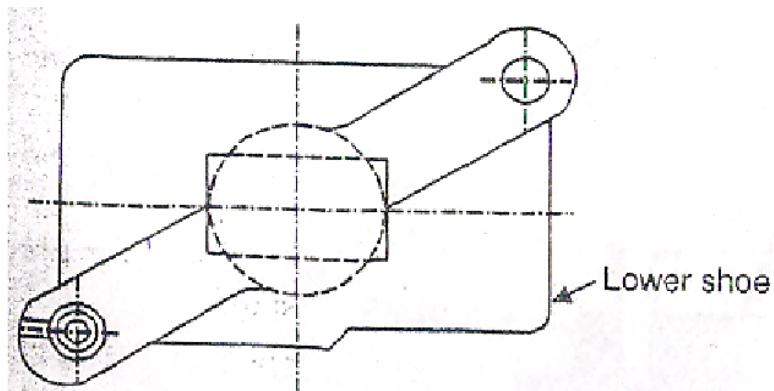


Fig. 2.9. Diagonal Post Design

d) **Four-guide post design:** In this design, Fig. 2.10, a guide post is positioned at each corner of the die-set. This design provides maximum rigidity and accuracy of alignment. This design is used for large, very complicated, or highly accurate stampings and for multi-station progressive dies.

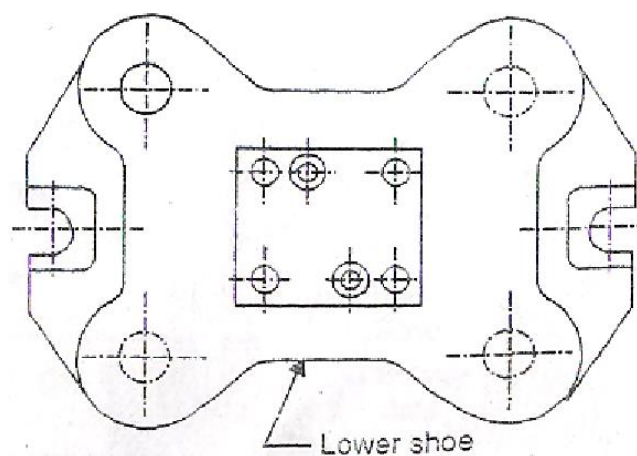


Fig. 2.10. Four-guide Post Design

PRESS SELECTION

The factors which should be considered while selecting a press for a given job, are: the overall work size, the stock thickness and material, kind of operation to be performed, power required and speed of operation.

For punching, blanking and trimming operations, usually the crank or eccentric type mechanical press is used. This is due to their small working strokes and high production rates. In these operations, there is sudden release of load at the end of the cutting stroke. This sudden release of load is not advisable in hydraulic presses. So, hydraulic presses are not preferred for these operations. If however these are inevitable, then some damping devices are incorporated in the press design. For coining and other squeezing operations, which require very large forces, knuckle joint mechanical press is ideally suited. Hydraulic presses, which are slower and more powerful, can also be used for these operations. Hydraulic presses are also better adapted to pressing, forming and drawing operations, which are basically slower processes.

PRESS WORKING TERMINOLOGY

A simple cutting die used for punching and blanking operations is shown in Fig. 2.11. Below, the definitions of the main components of the die and press are given:

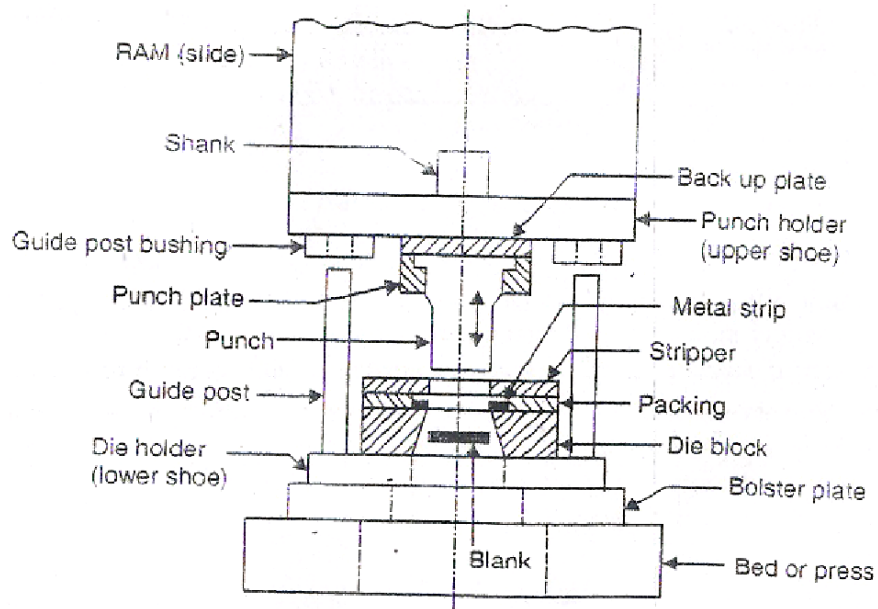


Fig. 2.11. A Simple Cutting Die.

- **Bed:** The bed is the lower part of a press frame that serves as a table to which a bolster plate is mounted.
- **Bolster plate:** This is a thick plate secured to the press bed, which is used for locating and supporting the die assembly. It is usually 5 to 12.5 cm thick.
- **Die set:** It is unit assembly which incorporates a lower and upper shoe, two or more guideposts and guidepost bushings.

- **Die:** The die may be defined as the female part of a complete tool for producing work in a press. It is also referred to a complete tool consisting of a pair of mating members for producing work in a press.
- **Die block:** It is a block or a plate which contains a die cavity.
- **Lower shoe:** The lower shoe of a die set is generally mounted on the bolster plate of a press. The die block is mounted on the lower shoe. Also, the guide posts are mounted in it.
- **Punch:** This is the male component of the die assembly, which is directly or indirectly moved by and fastened to the press ram or slide.
- **Upper shoe:** This is the upper part of the die set which contains guidepost bushings.
- **Punch plate:** The punch plate or punch retainer fits closely over the body of the punch and holds it in proper relative position.
- **Back up plate:** Back up plate or pressure plate is placed so that the intensity of pressure does not become excessive on punch holder. The plate distributes the pressure over a wide area and the intensity of pressure on the punch holder is reduced to avoid crushing.

- **Stripper:** It is a plate which is used to strip the metal strip from a cutting or non-cutting punch or die. It may also guide the sheet.
- **Knockout:** It is a connecting rod which is used to transmit motion from the main drive shaft to the press slide.
- **Shut height:** It is the distance from top of the bed to the bottom of the slide, with its stroke down and adjustment up.
- **Stroke:** The stroke of a press is the distance of ram movement from its up position to its down position. It is equal to twice the crankshaft throw or the eccentricity of the eccentric drive. It is constant for the crankshaft and eccentric drives but is variable on the hydraulic press.

Working of a cutting die (Fig. 2.11)

As is clear in the figure, the punch holder (upper shoe) is fastened directly to the ram of the punch press, and the die shoe (lower shoe) is fastened to the bolster plate of the press. Guide posts may be used to better align the punch holder with the die shoe. These main components (punch holder, die shoe and guide posts) constitute what is known as the die set. A die set can be had with two guide posts located at the rear of the die set (known as back posts), diagonally, one at the back and the other in the

front of the die set, or with four guide posts, one in each corner of the die set. The lower ends of the guide posts are press fitted into the die shoe. At the upper end, the guide posts have a slip fit with the guide bushings which are press fitted into the punch holder. With this, the guide posts have a free movement in the bushings. The punch is fastened to the punch holder and the die block is fastened to the die-shoe. The punch is aligned with the opening in the die-block. Since, both the punch and die block act as cutting tools, they are hardened.

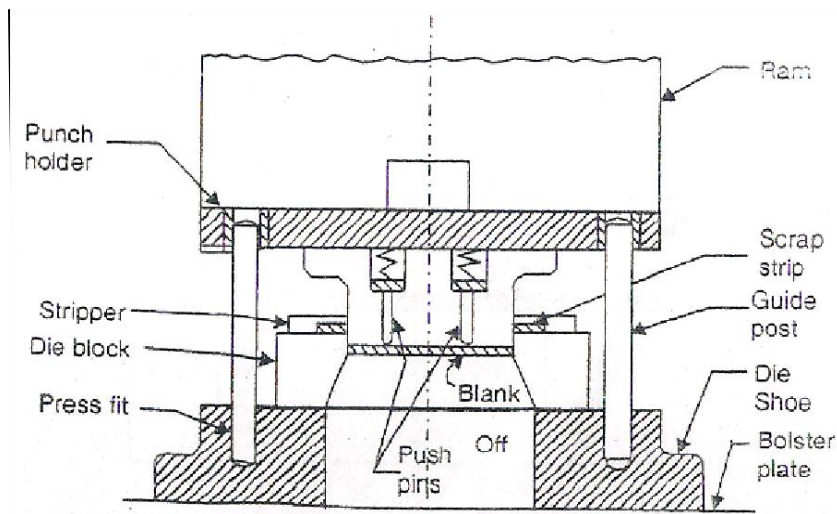


Fig. 2.12. A Cutting Die.

The cutting action takes place during the downward movement of the punch into the die block. After the cutting action, the elastic recovery

in the strip material takes place. Due to this, the size of the blank (cut portion from the strip) increases and that of the hole in the strip decreases. So, at the end of the cutting action, when the punch starts to move upwards the scrap strip clings to the punch and the blank gets clogged in the die-opening. To remove the scrap strip from the punch surface, a stripper is used. In Fig. 2.11., a simplest type of stripper is used. It strips off the scrap strip from the punch surface when the scrap strip strikes the bottom surface of the stripper during the upward movement of the punch. To avoid clinging of the blank in the die opening, the walls of the die-opening are tapered.

In addition to the two above mention troubles, the blank may also adhere to the face of the punch. This usually happens with thin blanks which have been treated with a lubricant. To help the blank free itself from the punch face, push-off pins are provided which are fitted into the punch body, Fig 2.12.

The various basic designs of push-off pins are shown in Fig. 2.13. The most popular design is a straight, round plunger with a sliding fit in the punch body, Fig. 2.13 (a). It is held in place by a light compression

spring and a set screw. If at the end of the punch body, a pilot is to be provided, then the push-off pin is located off-centre, Fig. 2.13 (b). When the space available for a compressing spring is insufficient, a flat spring may be used, Fig. 2.13 (c).

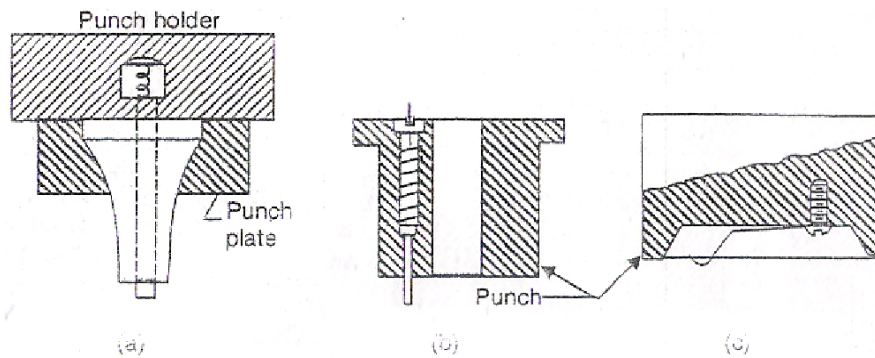
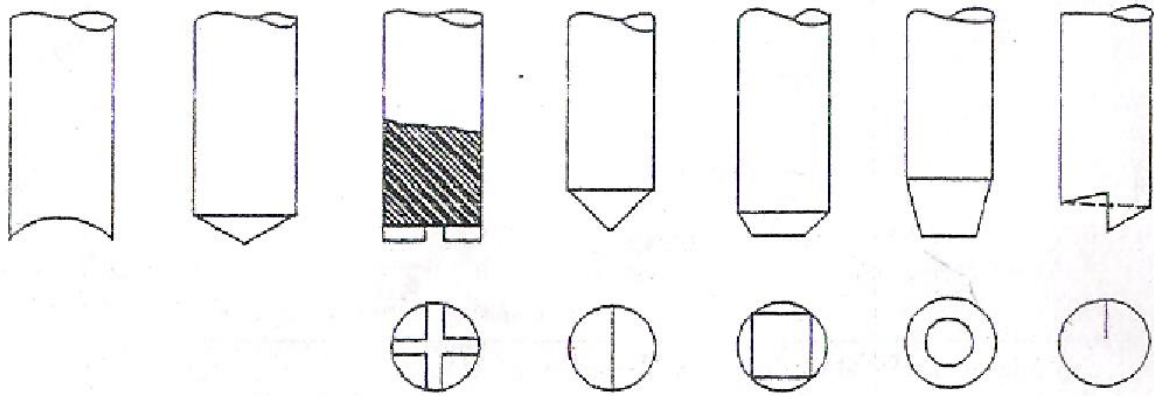


Fig. 2.13. Push-off Pins.

In the case of very small punches, no space is left to provide the push-off pin. In such cases, the sticking of the blank to the punch face may be avoided by changing the shape of the punch bottom, Fig. 2.14. With this, the oil seal or vacuum created by the oil film is broken and tendency of the blank to stick to the punch face is greatly decreased. The drawback of this method is that blank gets distorted. Due to this, this method is suitable for punching operation only.



2.14. Punch Face Contours to Prevent Blanks from Sticking to very small Punches.

TYPES OF DIES

This dies may be classified according to the type of press operation and according to the method of operation.

Type of Press Operation

According to this criterion, the dies may be classified as cutting dies and forming dies.

- **Cutting dies:** These dies are used to cut the metal. They utilize the cutting or shearing action. The common cutting dies are: blanking dies, piercing dies, perforating dies, notching, trimming, shaving and nibbling dies etc.

- **Forming dies:** These dies change the appearance of the blank without removing any stock. These dies include bending dies, drawing dies, squeezing dies etc.

Method of Operation

According to this criterion, the dies may be classified as: Single operation dies or simple dies, compound dies, combination dies, progressive dies, transfer dies, and multiple dies.

- **Simple dies:** Simple dies or single action dies perform single operation for each stroke of the press slide. The operation may be any of the operations listed under cutting or forming dies.
- **Compound dies:** In these dies, two or more operations may be performed at one station. Such dies are considered as cutting tools since, only cutting operations are carried out. Fig. 2.15 shows a simple compound die in which a washer is made by one stroke of the press. The washer is produced by simultaneous blanking and piercing operations. Compound dies are more accurate and economical in mass production as compared to single operation dies.

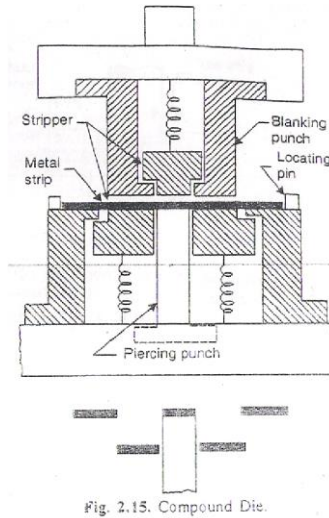


Fig. 2.15. Compound Die.

- Combination dies:** In this die also, more than one operation may be performed at one station. It differs from compound die in that in this die, a cutting operation is combined with a bending or drawing operation. Fig. 2.16 explains the working of a combination blank and draw die. The die ring which is mounted on the die-shoe is counter bored at the bottom to allow the flange of a pad to travel up and down. This pad is held flush with the face of the die by a spring. A drawing punch of required shape is fastened to the die shoe. The blanking punch is secured to the punch holder. A spring stripper strips the skeleton from the blanking punch. A knockout extending through the centre opening and through the punch stem ejects the part on the upstroke as it comes in contact with the knockout bar on the press. In operation, the

blank holding ring descends as the part is blanked then the drawing punch contacts and forces the blank into the drawing die which is made in the blanking punch.

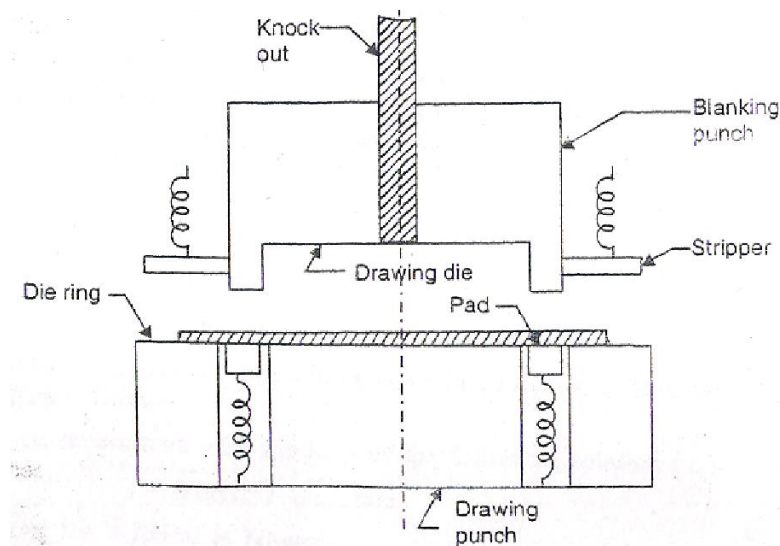


Fig. 2.16. Combination Die.

- **Progressive dies:** A progressive or follow on die has a series of stations. At each station, an operation is performed on a work piece during a stroke of the press. Between strokes, the piece in the metal strip is transferred to the next station. A finished work piece is made at each stroke of the press. A progressive die is shown in Fig. 2.17. While the piercing punch cuts a hole in the stock, the blanking punch blanks out a portion of stock, the blanking punch blanks out a portion of the metal in which a hole had been pierced at a previous station. Thus after

the first stroke, when previous station. Thus after the first stroke, when only a hole will be punched, each stroke of the press produces a finished washer.

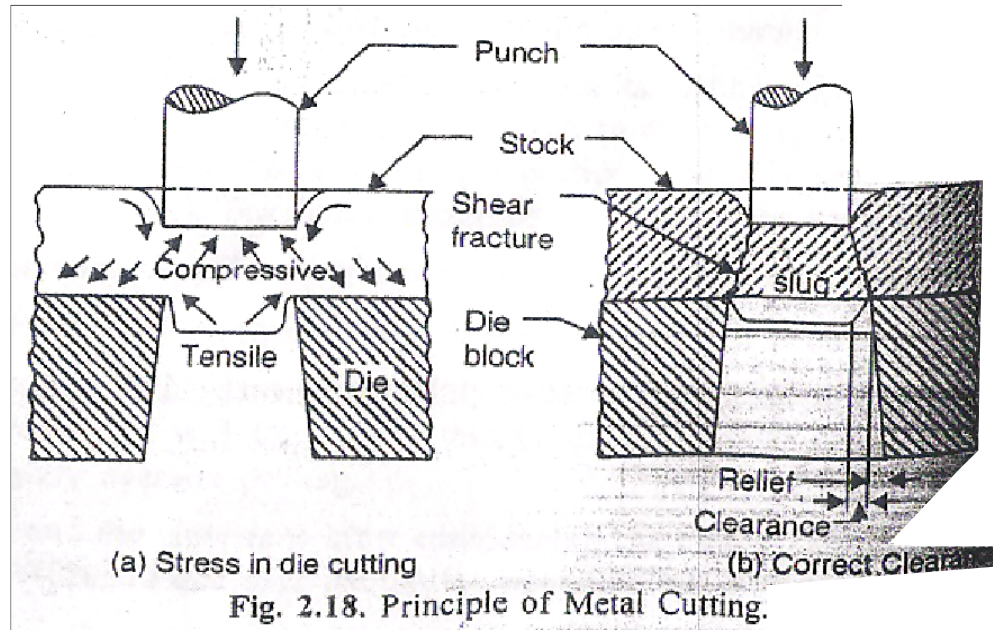
- **Transfer dies:** Unlike the progressive dies where the stock is fed progressively from one station to another, in transfer dies the already cut blanks are fed mechanically from station to station.
- **Multiple dies:** Multiple or gang dies produce two or more workpieces at each stroke of the press. A gang or number of simple dies and

punches are ganged together to produce two or more parts at each stroke of the press.

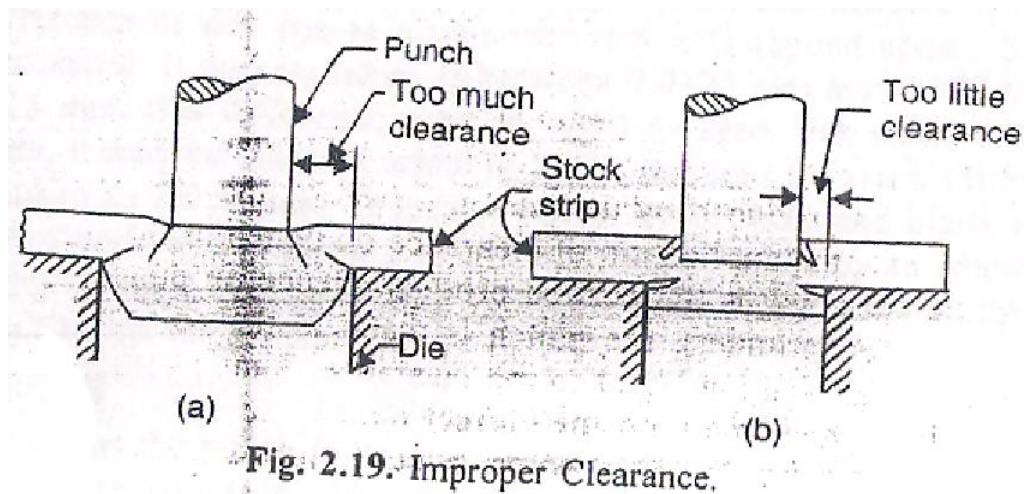
PRINCIPLE OF METAL CUTTING

The cutting of sheet metal in press work is a shearing process. The cutting action is explained with the help of Fig. 2.18. The punch is of the same shape as of the die opening except that it is smaller on each side by an amount known as ‘clearance’. As the punch touches the material and travels downwards, it pushes the material into the die opening, the material subjected to both tensile and compressive stresses as shown in Fig. 2.18 (a). Stresses will be highest at the edges of punch and die and the material will start cracking there. The various steps in the rupture or fracture of the material can be written as: stressing the material beyond its elastic limit, plastic deformation, reduction in area, fracturing starts in the reduced area and becomes complete. If the clearance between punch and die is correct, the cracks starting from the punch and die edges will meet and the rupture is complete as shown in Fig. 2.18 (b). If the clearance is

too large or too small, the cracks do not meet and a ragged edge results due to the material being dragged and torn through the die.



This is explain in Fig 2.19



CLEARANCE

As is clear in the previous pages, the die opening must be sufficiently larger than the punch to permit a clean fracture of the metal. This difference in dimensions between the mating members of a die set is called 'clearance'. This clearance is applied in the following manner:

- i. When the hole has to be held to size, i.e., the hole in the sheet metal is to be accurate (punching operation), and slug is to be discarded, the punch is made to the size of hole and the die opening size is obtained by adding clearance to the punch size, Fig. 2.20 (a).

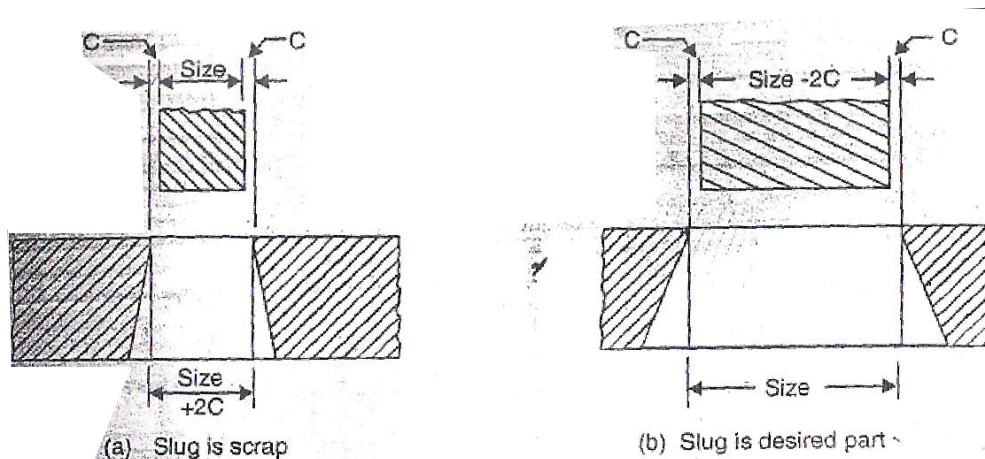


Fig. 2.20. Application of Clearance.

- ii. In blanking operation, where the slug or blank is the desired part and has to be held to the die opening size equals the blank size and the punch size is obtained by subtracting the clearance from the die-opening size, Fig. 2.20 (b).

The reason behind the application of clearance in the manner as given above, is explained below:

The diameter of the blank or punched hole is determined by the burnished area. On the blank, the burnished area is produced by the walls of the die. Therefore, the blank size will be equal to the size of die-opening (neglecting a slight expansion of the blank due to elastic recovery after the cutting operation is completed). Similarly, in punching operation, the burnished area in the hole is produced by the punch; therefore, the size of the hole will be the same as the punch. Therefore, the application of clearance on punch or die will depend on whether the punched hole or the cut blank is the desired product. Hence, in punching operation (where hole in the strip is the desired product), the punch is made to the correct hole size and the die opening is made oversize an amount equal to die clearance. Similarly, if the blank is the desired product, the die opening size is made to the correct blank size and the punch is made smaller an amount equal to die clearance. In other words, punch controls the hole size and die opening controls the blank size.

A section through blanking die is given in Fig. 2.21 showing clearance, land, straight and angular clearance.

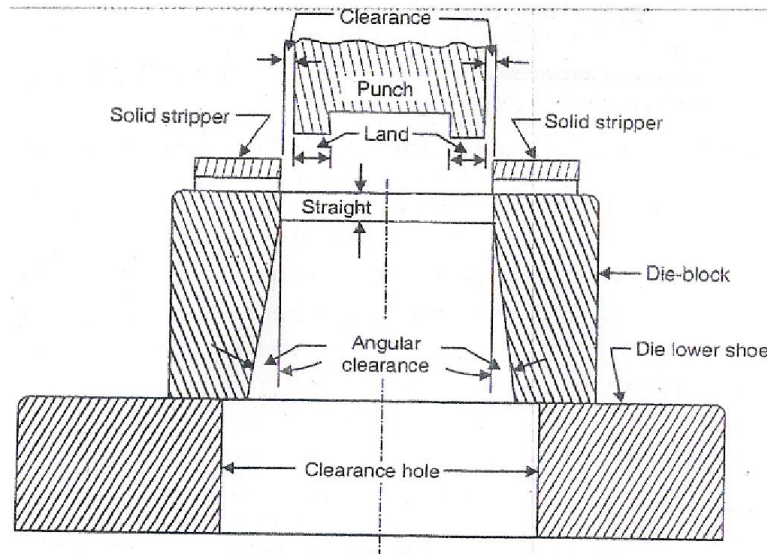


Fig. 2.21. Section through Blanking Die.

- **Land:** It is the flat (usually horizontal surface contiguous to the cutting of a die which is ground and reground to keep the cutting edges of the punch sharp.
- **Straight:** It is the surface of a cutting die between its cutting edge and the beginning of the angular clearance. This straight portion gives strength to the cutting surface of the die and also provides for sharpening of the die. This straight portion is usually kept at about 3 mm for all materials less than 2 mm thick. For thicker materials, it is taken to be equal to the metal thickness.

- **Angular clearance:** Angular clearance or relief is provided to enable the slug to clear the die. It is provided below the straight portion of the die surface. It is usually $\frac{1^\circ}{4}$ to $\frac{1^\circ}{2}$ per side but occasionally as high as 2° , depending mainly on stock thickness and frequency of sharpening for round die opening ($d \leq 3$ to 5 mm), instead of die angular clearance, die-clearance it maintained by slightly oversize drilling.

Punch and die clearance after considering the elastic recovery of the material

It has already been explained that after the cutting operation has been completed, elastic recovery of the strip material takes place. In blanking operation, after the release of blanking pressure, the blank expands slightly. The blanked part is thus actually larger than the die opening that has produced it. Similarly, in punching operation, after the strip is stripped off the punch, the material recovers and the hole contracts. Thus, the hole is actually smaller than the size of the punch which produced it. This difference in size due to elastic recovery will depend upon: blank size, stock thickness and stock material. It may be taken as

between 0.0125 mm and 0.075 mm. If the stock thickness is up to 0.25 mm, this difference may be taken as zero. For stock thickness between 0.25 mm and 0.75 mm, it may be taken as equal to 0.025 mm and for stock thickness more than 0.75 mm, it may be taken as 0.050 mm. Thus to produce correct hole and blank sizes, the punch size should be increase and the die opening size should be decreased by an amount as explained above. Or, the swelling/shrinkage expected due to elastic deformation may be taken as $= (0.003 \text{ to } 0.01) \times t$; for softer to harder stocks.

CUTTING FORCES

In cutting operation, as the punch in its downward movement enters the material, it need not penetrate the thickness of the stock in order to affect complete rupture of the part. The distance which the punch enters into the work material to cause rupture to take place is called ‘penetration’ and is usually given as the percentage of the stock thickness.

The per cent penetration depends on the material being cut and also on the stock thickness. When a hard and strong material is being cut, very little penetration of the punch is necessary to cause fracture. With softer

materials, the penetration will be greater. For example, for soft aluminium, it is 60% of 't'; for 0.15% carbon steel annealed, it is 38% of 't'; and only 24% of 't' for 0.5 per cent carbon steel annealed. The percentage penetration also depends upon the stock thickness, being smaller for thicker sheets and greater for thinner sheets, as shown in Table 2.1. i.e., it is inversely proportional to stock thickness.

Table 2.1 Penetration

Stock Thickness t.mm	25	20	15	12.5	10	8	6	5	3	2.5	1.6	Below 1.6
Penetration % of t	25	31	34	37	44	47	50	56	62	67	70	80

The maximum force $F_{\max}$ in Newton's needed to cut a material is equal to the area to be sheared times the shearing strength, T_s in N/mm^2 for the material. For a circular blank of diameter D mm and of thickness t , mm, the cutting force will be given as,

$$F_{\max} = \pi D t T_s = P \cdot t \cdot T_s \dots \dots \dots (2.1)$$

Where 'P' is the perimeter of the section to be blanked.

For rectangular blanks with length L and width b , it is

$$F_{\max} = 2 (L + b) t.T_s \dots \dots \dots (2.2)$$

The shear strengths of the various metals are given in Table 2.2

Table 2.2. Shear Strengths of Various Metals

<i>Metal</i>	<i>T_s in N/mm^2</i>
Carbon steels:	
0.10%C	245 to 311
0.20%C	308 to 385
0.30%	364 to 469
High strength low-alloy steels	315 to 446
Silicon steels	420 to 490
Stainless steels	399 to 903
Aluminium alloys	49 to 322
Copper and bronze	154 to 490
Lead alloys	12.8 to 41
Magnesium alloys	119 to 203

Nickel alloys	245 to 812
Tin alloys	20.5 to 77.7
Titanium alloys	420 to 490
Zinc alloys	98.0 to 266

To allow for energy lost in machine friction and in pushing slugs through the die, etc.,

Press capacity will be taken as $= F_{\max} \times C$,

Where the factor C is

= 1.1 to 1.5 for normal to narrow profiles

= 1.25 to 1.75 for $d/t < 2$

METHODS OF REDUCING CUTTING FORCES

For calculating the cutting forces in the last article, it has been assumed that the bottom of the punch and the top of the die block lie in parallel planes and that the blank is severed from the sheet metal by shearing it simultaneously along the whole perimeter. This process characterized by very high punch forces exerted over a very short time, resulting in shock or impulse conditions. It is usual, however, to reduce cutting forces and to smooth out the shock impact of heavy loads. This is

achieved by arranging for a gradual cut instead of sudden cut of the stock.

For this, two methods are generally used:

1. **Shear:** The working faces of the punch or die are ground off so that these don't remain parallel to the horizontal plane but are inclined to it. This angle of inclination is called 'shear'. This has the effect of reducing the area in shear at any one time and the maximum force is much or less. It may be reduced by as much as 50%. This is made clear in Fig. 2.22, which shows the relation of cutting forces to amount to shear.

In Fig. 2.22 (a), the shear is zero, i.e., the cutting edges are parallel. The material is cut at once on the entire perimeter resulting in maximum load. The force diagram shows a steep rise at maximum load and then sudden load release sometimes severe on both press and dies, as the cut is completed. In Fig. 2.22 (b), the face of the punch is ground off so that $\text{shear} = t/3$. The cutting action will start at the leading edge of the punch and then it will gradually spread to the rest of the punch. With this, only a part of the punch would be cutting at any one instant. While the maximum force will decrease, the energy needed to complete the cut is unchanged. So, the punch travel in this case will be more than in case (a).

In case 2.22 (c), shear = $t/1$. When the leading edge has travelled through the stock a distance 't', the trailing edge will start making contact with the material. Maximum force would be at this position of the punch and since the cut is complete, the maximum load would be about half of that when shear is zero. The punch travel will be still greater.

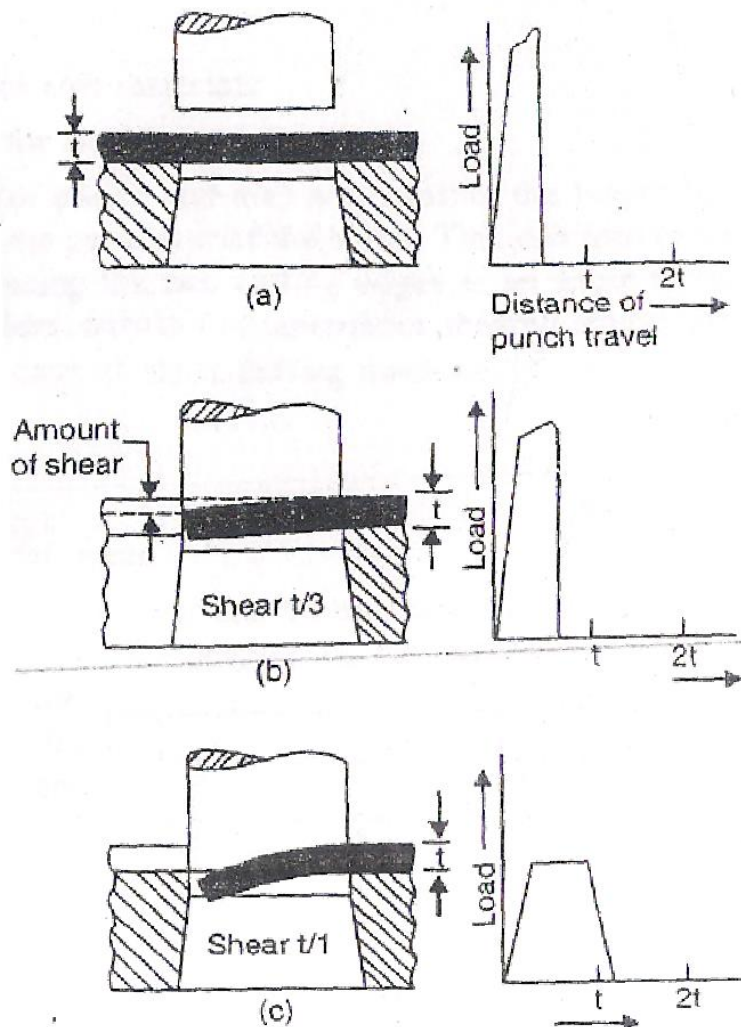


Fig. 2.22. Effect of Shear on Cutting Force.

The provision of shear distorts the material being cut. When shear is on the face of the punch, the blank cannot be flat and when shear is on the die, the piercing cannot be flat. So, for blanking operation, shear is provided on the die face and for punching or piercing operation shear is provided on the punch face. Fig. 2.23 shows the various methods of applying shear on the punch and die face. Wherever possible, double shear should be used so that the two shear faces neutralize the side thrusts which each sets up.

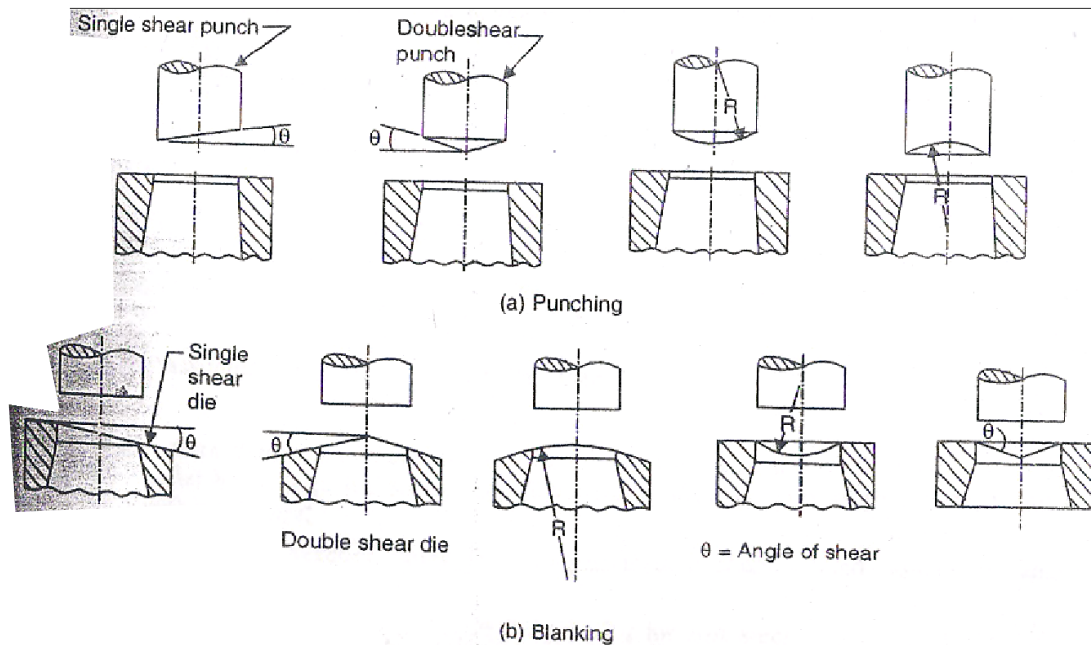


Fig. 2.23. Application of Shear.

It is clear from the above analysis that it is advisable to provide shear on punches and dies. The amount of shear to be applied is a matter of

comprise. If the shear is quite big, say $2t$ or $3t$. However, the shear must be at least equal to the percentage penetration.

The effect of shear on cutting force can be determined by comparing the work done with shear and without shear. From Fig. 2.22, it is clear that,

$$\begin{aligned}\text{Work done} &= \text{Area under curve} \\ &= (\text{maximum punch force}) \times (\text{punch travel})\end{aligned}$$

If the shear on the cutting tools is zero, then

$$\text{Punch travel} = (\% \text{ penetration}) \times (\text{material thickness})$$

$$\text{Work done} = F_{\max} \times K \times t$$

(2.3)

Where K = percentage of penetration required to cause rupture

If shear is provided on punch or die, then

$$\text{Punch travel} = K \times t + \text{Amount of shear in cm} = K \times t + l$$

l

Where l = total inclination or shear of punch or die, in cm

$$\text{Work done} = F \times (K \times t + l) \quad (2.4)$$

Where, F , is the actual cutting force and it will be less than, $F_{\max}$.

The relations (2.3) and (2.4) are based on the assumption that punch force $F_{\max}$ and F , sustained during complete penetration of work material, which is not true. So, the analysis is on an approximate one.

Since work done remains the same, therefore, comparing equations (2.3 and 2.4), we

$$l = \frac{(F_{\max} - F) \times K \times t}{F} \quad \text{and} \quad F = \frac{F_{\max} \times K \times t}{K \times t + l}$$

The above is true for single or double shear.

It is clear that the provision of shear increases the punch travel to complete severance of a material. This results in a smoother cutting action and cleaner edges with a higher degree of squareness. However, the initial die cost increases and there is also greater chance in dimensional error and variation, since the dimensions of the cut tend to increase in proportion to the angle of shear. The shear angle chosen should provide a change in punch length of about 1 to 2 times stock thickness. In piercing, the direction of shear angles must be such that the proceeds from the outer extremities of the contour towards the centre. This avoids stretching material when it is cut free.

The application of shear reduces the cutting force according to the penetration before breakthrough. The ratio of penetration to the shear gives the reduction factor. Thus, a simple relation can be

$$F = \frac{Kt}{l} \times F_{\max} \text{ with } l > Kt \text{ or } F = P \cdot t \cdot t_s \times k$$

$$\begin{aligned} \text{The factor } k \text{ is} &= 0.2 \text{ to } 0.6 \text{ if shear } l = t \\ &= 0.2 \text{ to } 0.4 \text{ if shear } l = 2t \end{aligned}$$

As already noted, when the cutting edges (of punch and die) are parallel, the length to be cut is the entire length of the contour cut, that is, the perimeter of the blank. This can lead to very high cutting forces. These can be reduced by placing the two cutting edges at an angle to each other (called shear or rake) as discussed above. Here, only the instantaneous sheared length needs to be considered, which will be given as in the case of sheet cutting machine,

$$= \frac{t}{\sin \alpha}$$

Where: α = angle of shear.

2. Staggering of punches: As an effect similar to shear can be obtained by staggering two or more punches that all operate in one stroke of press.

For staggering, the punches are arranged so that one does not enter the material until the one before it has penetrated through. In this manner, the cutting load may be reduced approximately 50 per cent, Fig. 2.24.

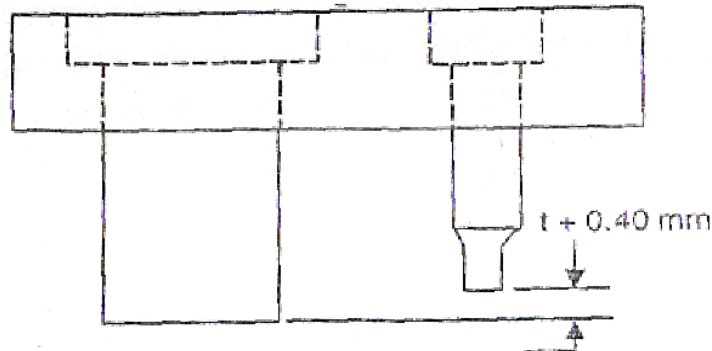


Fig. 2.24. Staggering of Punches.

MINIMUM DIAMETER OF PIERCING

To punch a hole without the failure of the punch, the compressive strength of the punch should be at least equal to the force necessary to fracture the material. The hole which will be punched under this condition will be the smallest. If t_s is the shear strength of the work material α_c is the compressive strength of the punch material, d is the smallest hole diameter to be pierced and t is the thickness, then,

$$\text{Piercing pressure} = t_s \cdot \pi d \cdot t$$

$$\text{If it is assumed that,} \quad = \alpha_c \cdot \frac{\pi d^2}{4}$$

$$t_s \cdot \pi d \cdot t = \alpha_c \cdot \frac{\pi}{4} d^2$$

$$d = \frac{4t_s}{\alpha_c} \cdot t$$

If it is assumed that, $\alpha_c = 2t_s$ then,

$$d = 2t$$

i.e., the minimum diameter of hole that can be punched is twice the stock thickness. In actual practices, small holes than this are pierced successfully, showing to what extent the punch may be loaded without failure.

Minimum size of punched holes depending upon their shape (round, square, rectangular, oval).

BLANKING DIE DESIGN

Types of Blanking Die

There are two general types of blanking dies: the drop through and the inverted.

Drop-through die: In this die, the die block assembly is mounted on the bolster plate or the press bed and the punch assembly on the press slide, fig.2.11. The blank drops of its own weight through the die opening

and the clearance hole provided in the bolster plate and press bed. This design is economical to build and maintain and is fast in working. However, this design is not suitable under the following condition:

1. When the blank is too thin and fragile to be dropped very far
2. When the blank is too heavy to be dropped for any appreciable distance.
3. When the blank is too awkward to be removed from below the press.
4. When the blank is larger than the press bed opening

Inverted type dies: In this design, the punch becomes the lower stationary part and the die is mounted on the ram. This type is somewhat more complicated, more costly and slower in operation. The scrap disposal is much easier but removal of blank from the die opening is troublesome and a device is needed to knock the blank out of the die opening. This type is widely used where the blank is large or heavy.

Fig. 2.25 shows an inverted type Blanking die. During the downward stroke of the ram blank is cut from the strip. The blank and the shedder (spring loaded knockout) are forced into the die opening, which compresses a spring in the die opening. At the same time, a compression spring attached to the stripper is compressed. On the upstroke of the ram, the stripper strips scrap off the punch and the shedder is forced out of the opening. The blank falls down or is blown out the rear of the press. Usually, the press is inclined to aid the discharge operation.

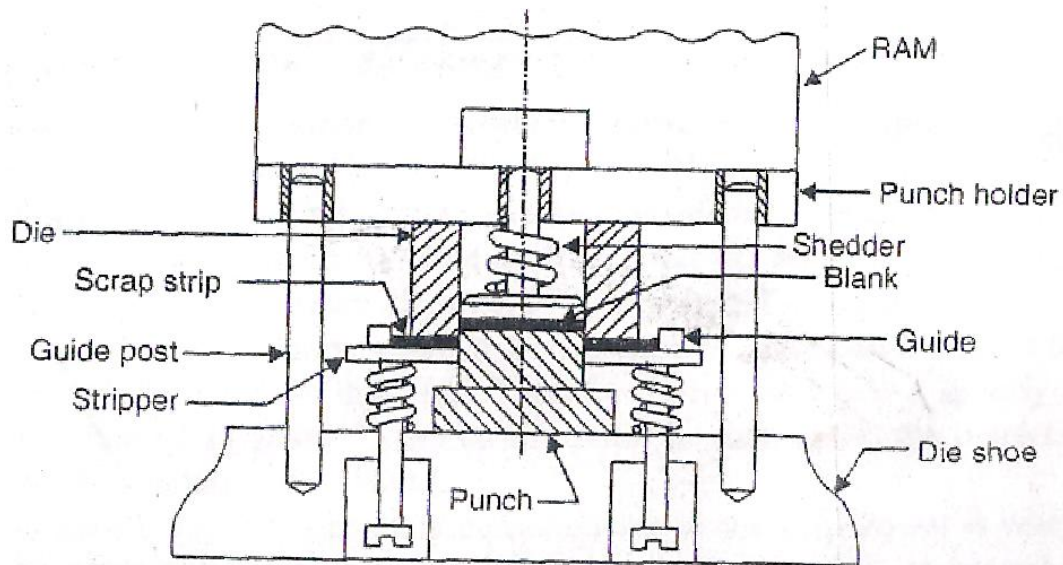


Fig. 2.25. Inverted Blanking Die.

Strip Layout

In the design of a blanking die set the first step is to prepare blank layout that is to layout the position of the work pieces in the strip and their orientation respect to one another. This is called 'strip layout'. The factors will influence the layout are:

1. Economy of material.
2. Direction of material grain or Fibred.
3. Strip or coiled stroke.
4. Direction of burr.
5. Press used
6. Production required
7. Die cost.

1. Economy of Material: In fig. 2.26, the different ways of arranging to blank the given work piece are shown. The arrangement in fig. 2.26 (a) can be worked at single row, single pass with a single punch. For arrangement of fig. 2.26 (b), the strip would either have to be fed twice, once for each row, or double blanking will have to be employed. The per cent material utilization may be increased somewhat by the arrangement of fig. 2.26 (c) shows a single row, double pass strip. This is called stock nesting". Here the strip will have to be passed through the dies once turned over, and passed through dies a second time. Nesting costing considerably reduces the scrap. However, the strip layout with maximum material saving may not be the best strip layout, as the die construction may become more complex which would offset the savings due to material economy unless a large number of parts are to be produced.

Another important consideration in strip layout is the distance between the nearest point of blanks and between blanks and the edges, of

the strip. To prevent the scrap from twisting and wedging between the punch and the die, this distance must increase with material thickness. A general rule of thumb is to keep this distance, called as web, at least 1.5 times the material thickness. However, other factors such as strip thickness, hardness of material, type of operation. Shape of blank etc. may allow the web to be thinner. The various terms connected with strip layout are shown in fig.2.27.

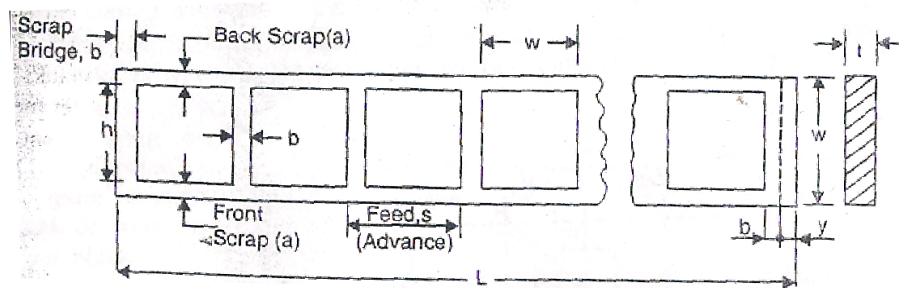


Fig. 2.27. Strip Layout.

The distance between the blank and edge of strip, known as back scrap (font scrap) may be determined by the equation.

$$a = t + 0.015 h \quad \dots(2.8)$$

The distance between successive blanks and also the scrap bridge, b , is given table 2.3 in general, softer material require larger spacing. The feed or advance or the length of one piece of stock needed to produce one blank is

$$s = w + b(2.9)$$

The number of banks which can be produced form one length of stock can be found out as.

$$N = \frac{L-b}{s}(2.10)$$

Material thickness	B
Mm	Mm

0.8	0.8
0.8 to 3.2	t
Over 3.2	32

The scrap remaining at the end of one length of strip may be calculated from,

$$y, = L - (Ns + b)(2.11)$$

Measure of material utilization: it is defined as,

$$n, = \frac{\text{area of blank to be cut}}{\text{area of material available}}$$

$$= \frac{B}{A} \times 100$$

$$\%scrap = \frac{A - B}{A} \times 100$$

Now area of material available per blank= feed or advance× Stock width Feed is also called “progression”.

$$\text{For fig. 2.27, \% utilization, } n = \frac{w \times h}{(w+b)(h+2a)},$$

For a round blank of diameter d,

N should not be less than 70% for economical working, however, for complicated shapes blanks, n can be less than 70%.

2. Direction of Material Grain: This factor is to be considered if the cut blank have undergone any subsequent operation, such as, bending or deep drawing. When the sheet metal strip is rolled in the mill, a fiber is produced in the direction strip length. During the subsequent bending operation on the blank, to obtain maximum strength from bent parts, the bend should be made across the strip, or at an angle of 90 to the fiber. Therefore some part print specify that the fiber is to run in the direction of an arrow shown on the print. In such cases, the blanks cannot be tipped or rotated to just any position desired.

3. Strip or Coiled Stock: Another important consideration in the strip layout is whether the stock used will be in the form of strip or coil. Whereas, the stock strip may passed through only once.

When coiled stock is used, recoiling reel. A stock straighter and feed roll is necessary. Thus coiled stock is used when

- a) Production is high.
- b) Thinner metal sheets are employed.
- c) Stock need to be passed through the die only once.

Strip stock is used when

- i. Production is low.
- ii. Thicker sheet metals are used
- iii. The stock needed to be passed through the die more once

Here the stock feed is manual.

- 4. Direction of Burr:** When sheet metal is cut in a die, a burr is produced on the die side of the scrap strip and on the punch side of the blank. If the burr has to be on the hidden side, then the expensive operation of removing the burr need not be done. For this a note is often placed on the part drawing which reads “burr down”. To control the position of the burr may limit the stock-layout arrangement. Scrap may not be reduced to a minimum.
- 5. Press Used:** During production planning, a press has been assigned to the operation and the die. Therefore, the stock layout has to be such that it allows the die to be designed within the press capacity. Shear may have to be provided on punch or die, to limit the maximum cutting force within the press capacity. Another factor is the bed area of the press. The third factor is to have the cutting forces of the die evenly balanced around the centre line of the press ram.
- 6. Production Required:** The following guide lines may be followed when production is the main consideration.
- a) Low production-thin material
 - i. Strip stock and a single-pass layout.
 - ii. Cutting of one or more blanks at a time
 - b) Low production-thick material
 - i. Strip stock and a single or double-pass layout.
 - ii. Cutting of one blank at a time.
 - c) High production-thin material

- i. Coiled stock and a single-pass layout.
 - ii. Cutting of one or more blanks at a time.
- d) High production-thick material.
- i. Strip stock, and a single-or double –pass layout.
 - ii. Cutting more than one at a time.
7. **Die-Cost:** Die-cost will be higher for:
- i. Higher productions.
 - ii. Cutting more than one at a time, particularly when cutting extremely complicated blank shapes, or when cutting extremely accurate blank sizes.

However, for simple round or square-edged blanks, multiple cutting at one time is often practical. Also, double –pass dies are less expensive than cutting two at a often practical. Also, double-pass dies are less expensive than cutting two at a time so the designer has to decide while making the stock layout, as to which is preferred: more operator time per blank or more machine time per blank.

Die-Block

The die block is the female half of the two mated tools which carry the cutting edges. It is subjected to extreme pressures and wear conditions. Hence the die block is made of a superior quality of tool steel. A simple layout of the die bock is shown in fig. 2.28. The die block may be of solid

or sectional construction depending on the size and contour of the die opening is small and its contour is simple.

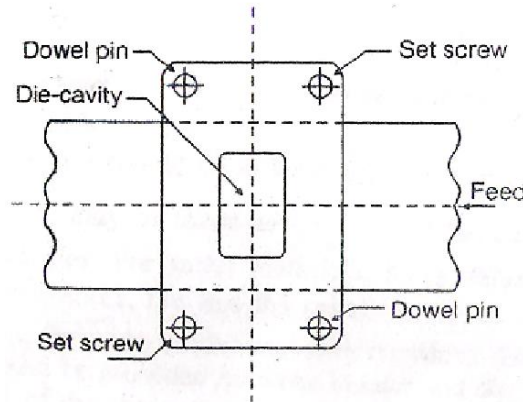


Fig. 2.28. Die Block Layout.

A solid die block is the choice. Sectional die are made up of accurately ground matching components which may be assembled together easily. Sectional dies have the following advantages.

1. Dies with long and complicated contours can be broken up into sections of simple geometrical forms which can easily and economically be machined.
2. Building the die block in sections eliminates heat treatment, treating, distortion and cracking as the separate pieces handled are more uniform in cross-section.
3. Grinding of the section is more easily done.
4. Maintenance of die is simple and less costly, because if one section happens to crack, only that particular section will have to be replaced.

Sectional Die has the following disadvantages:

1. This design is not suitable when the stock thickness is quite large and there is great side pressure in blanking.
2. If the bed and press slide are not absolutely parallel, proper an alignment between the punch and the sectional die will be extremely difficult to maintain.

For the high production of accurately shave die products of complicated shapes die have definitely an edge over solid dies. However, the available of modern tooling equipment by which punch and die can die can be made in one cut for example by contour sawing method has extended the range of solid die construction. Generally, high carbon, high chromium steels are used for die section which are hardened and ground. The die block is either mounted on or held in a die shoe, shoe, as shown in fig. 2.29.

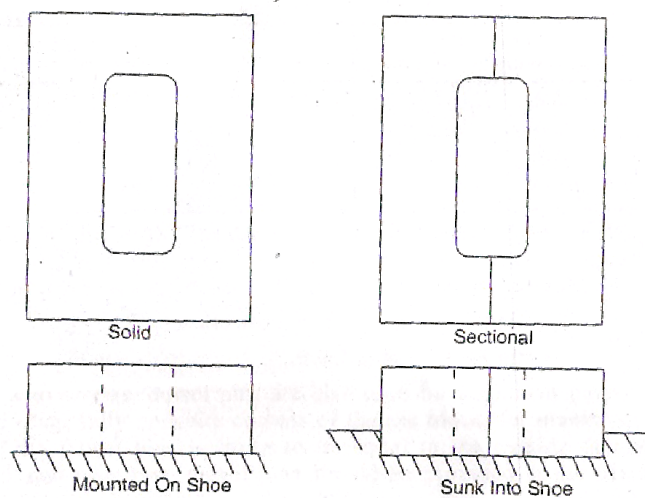


Fig. 2.29. Fixing of Die Block to Die Shoe.

Die Block Thickness

The minimum thickness of die block depends upon the strength required to resist the force, and it will depend upon the type and thickness of the material being cut. The determination of die block thickness is usually based on experience and thumb rules.

According to one “rule of thumb” the die thickness may be obtained as follows:

- i. Die thickness=19 mm, for blank perimeter> 75 mm
- ii. Die thickness=25 mm, for blank perimeter=75mm to 250mm
- iii. Die thickness=31 mm, for blank perimeter=250mm.

This rule of thumb is for die blocks made of too steel.

The die thickness may also be obtained from table 2.4 in which it is given as a function of stock thickness and quantity of production. Allowances for polishing must be added to the values.

When the parts are made of lamination stock or other extremely hard of tough materials the thickness given in table 2.4 can be increased by 3 mm, however. For Bakelite brass or similar soft materials, it can be reduced by 3 mm. care must be taken not to make the die block unnecessary thick as it adds to the cost of manufacture and results in a excessive shut height.

Table 2.4.Die Block Thickness.

<i>Stock thickness mm</i>	<i>Production quantity (million)</i>	

	5	25	50	100	500	<i>Unlimited</i>	<i>Die block thickness mm</i>
0.375	9.375	12.5	15.625	19	25	28	
0.775	12.5	15.625	19	25	28	28	
2.325	15.625	19	25	28	31	34	
3.125	19	25	28	31	31	34	
4.675	25	28	28	31	34	37.5	
9.25	31	31	31	34	37.5	37.5	

Or die-block thickness can be taken as

$$T = 3\sqrt{f \text{ cm}; f \text{ is in tonnes}}$$

$$T = 7.5 \text{ mm and } 10.5 \text{ mm}$$

For die-surface are above 3200mm².

Die Opening

A section of the block is shown in fig.2.21. The side walls of the die block opening should be provided with sufficient relief, fig. 2.18a or taper so that the blank drops clear through. This taper can either start from the top surface itself or after a straight land from the surface of the die, fig. 2.2. where the filling and grinding of the die is done by machine. The fully tapered cavity is quicker to produce and is therefore cheaper. But if the die has to be finished on bench, the straight land is easier to file and sandstone.

Soft metal such as copper brass and aluminum tend to swell more rapidly after being cut. This may be stressed in compression. So, for such

metals, the die cavity should be fully tapered. For steel, the taper or relief should start after the straight land from the die surface. The chief advantage of the straight and is that the original dimensions of the die are retained after repeated regrinding. With fully tapered die cavity design, the die opening size increase, however, is very negligible to cause any appreciable effect on blank dimensions.

As already mentioned, the straight land should be at least equal to stock thickness and not less than 3 mm. The relief angle or taken as for small dies, 1° to 2° for average die 2° to 3° on large dies. For soft materials, these values can be slightly increased. Also the close the blank tolerances, the less the relief.

The nominal size and outline of the die cavity is identical with the blank dimensions. At least a distance equal to stock thickness should be provided between blanks and the edge of the strip. The die wall thickness from the edge of the die opening to the die block border, that is, the distance A [fig. 2.29 (a)]

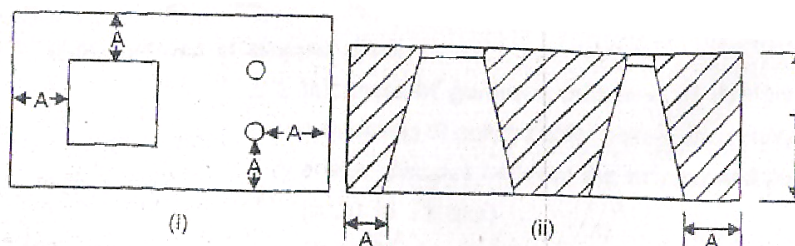


Fig. 2.29 (a)

After that, the die thickness T must be checked. According to an empirical rule, the section area $A \times T$, [fig. 2.29_a (ii)] must bear a certain

minimum relationship to the maximum cutting force for a die put on a flat base. This given in table 2.4 (a) if the die thickness as calculated above, does not give sufficient area for the critical distance must be increased accordingly.

According to rule of thumb” there should be a minimum of 32 mm margin around opening of the die block (distance A). in general on very thin materials, 12.7mm die block thickness will be sufficient but except for temporary tools, finished thickness (include grinding allowance of 2.5 to 5mm) is seldom less than 22 mm, which allows for blind holes. Sharp corners in the contour may lead to cracking in heat treatment, and so require great wall thickness at such points.

<i>Maximum cutting force kN</i>	<i>Area between die opening Border(cm²)</i>
200	3.25
500	6.50
750	9.75
1000	13.00

Fastening of Die Block

The die block is secured is either to the die shoe or bolster plate size of screws and bolts employed is usually to calculated. They are usually chosen person judgment and practice experience. According to one experience. According to one empirical rule.

$$\text{Screw diameter} = 0.5 t, \text{ for } T \leq 19 \text{ mm.}$$

$$=0.4 t, \text{ for } T \geq 19 \text{ mm.}$$

Along with screws, dowel pins are also used for alignment purposes. They are usually located near diagonally opposite corners of the die block, for maximum locating effect diameter of the dowel pins is taken to be equal to the outside diameter of the faster screws. Two, and only two, dowel pins should be provided in the die block for permanent positioning.

Dowel Pins

Any misalignment of die details may cause severe damage. There details that must be located accurately are held in such a position by dowels. Two dowel necessary for one detail to achieve are complete location. Most dowels used in the detail and the mounting plate or keeping the dowel diameters same as the screws facilities the drilling and reaming dowel holes.

The position of the dowel pins is so selected as to make through holes for them fig. 2.30 otherwise the removal of the pins during disassembly will be too difficult.

If though holes not available tapes pins are driven, the holes for them being machined by a taper reamer.

Whenever possible, threading hardened components should be avoided. For example, the threads should be cut in the die-shoe (which is made of c-steel and is not hardened) and die block should be bored to

accept the cap screws. The effective thread depth of the screws should be 1.5 to 2 times the screw diameter.

- i. On die blocks up to 18cm square use two 10mm dowel pins.
- ii. On section up to 25 square, use three screws and two dowel pins
- iii. For blanking heavy stock, use screw and dowel of 12.7 times diameters, counter bore the cap screws about 3 mm deeper than usual to compensate for die sharpening.

Screws and dowels are preferably located about 1.5 to 2 times their diameter from the outer edge or the blanking contour, min. permissible centre distance between the tapped hole and the dowels are $\text{pin} \geq 2 \times \text{screw diameter}$.

The diameter of fastening screws can also be found by calculations. The basis for the computation is the stripping force which depends upon many factors, but be taken as 10 percent of cutting force: with this force the screw which fasten the stripper to the die plate or secure the punch holder to the punch plate can be easily calculated practically, the same values are then taken for screws which the die block to the press table.

$$\text{Stripping force, fig.} = 0.1 \times p \times t \times \tau = \pi d_c \times n \times \sigma$$

Where d_c is the core diameter of screw, σ is the safe tensile stress of screw material and n is the number of screws usually an even number; 2 for small dies, 4 for medium dies and 6, 8 or more for dies.

Not. A minimum of 16 mm should be provided between punch and blocks guide pins and bushings to allow clearance for the grinding wheel

when this is not possible removable guide posts may be used for easy access to dies for maintenance purposes the die holder or the lower shoe should be at least 6.5 mm larger all around the die block.

Punch

The punch must be a perfect mate to the die block opening the size of the working surface of punch is obtained by subtracting the total clearance from the desired size of the blank. A already mentioned shear is provided on the die surface for blanking operation. The punch is usually is provide with wide flange or shoulder to facilitate mount and prevent its deflection under load. The minimum length of punch should be such that it extends far enough in to the die block opening to ensure complete shearing of the blank. The punch length must also provide for the anticipated number of regrinds. The maximum length of the punch can be calculated from the formula, from the consideration of an axially loaded street.

$$L = \frac{\pi d}{8} \left(L = \frac{e}{\tau_s} \frac{d}{t} \right) \quad \dots(2.13)$$

Where E is the modulus of elasticity d is punch diameter or maximum permissible free length of circular punches to be safe against bucking can also be determine as.

$$L_{\max} = 7.5 \frac{d^3}{1} \text{ units are mm}$$

Total length of punch guided in stripper plate,

$$L \geq \text{thickness of guides} + \text{thickness of stripper plate}$$

+ thickness of punch holder plate + sharpening allowance

+ min. distance between the stripper and punch holder

(≈ 20 to 25 mm)

L = 60 to 85 mm

Generally.

Punches are made of good grade tool steel, hardened and ground. The hardness recommended is Rockwell C 60 to 62.

Punches with unguided length of more than 100 mm

are avoided in sharpening allowance of 6 to 12.5 mm is provided in punch length.

The strength of the punch is checked for the weakest cross-sectional area by the formula,

$$\sigma_{\text{comp}} = \frac{F_{\text{max}}}{A} \leq 1\sigma_1 \text{ permissible}$$

Where A = cross-sectional area of punch at its weakest section.

Back Up Plate

For small punches, back up plates or pressure plates are often provided between the punch plate and punch holder (fig. 2.31). The punch plate or punch retainer fits closely over the body of the punch and holds it in proper relative position. It is attached to the punch holder. The back up plate is provided as a base and insurance against punch deflection when punch does not have a flange or shoulder and prevent the hardened punch from being pushed into the softer punch holder thus becoming loose.

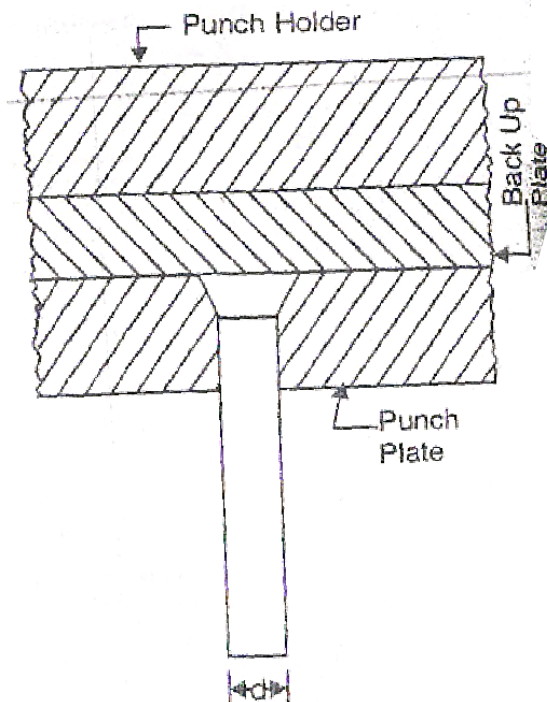


Fig. 2.31. Back up Plate.

The main criteria whether a back up plate should be provided or not is the unit compressive stress on the punch given as,

$$P = f/A$$

Where, A is the cross-sectional area of the punch.

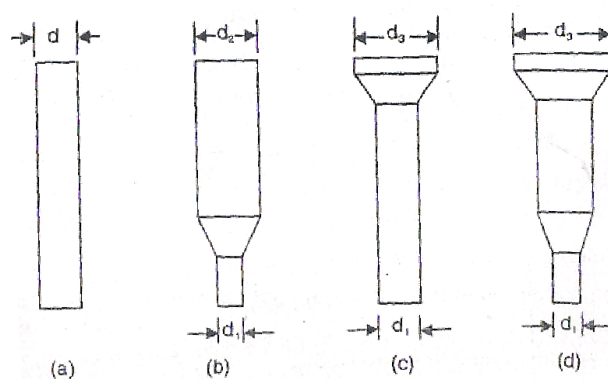


Fig. 2.32. Punch Construction.

METHODS OF HOLDING PUNCHES

Punch holder does not present any problems. Being relating or securing of a blanking punch holder does not present any problems. Being relatively bigger they are made with flanges that are doweled in to position and directly fastened to the punch holder by screws without the use of punch plate and sometimes without even a back up plate (fig. 2.33). when used, the thickness of punch plate should be plate 1.5 times the diameter.

For press fitted punches, a fit H/p , is to be maintained.

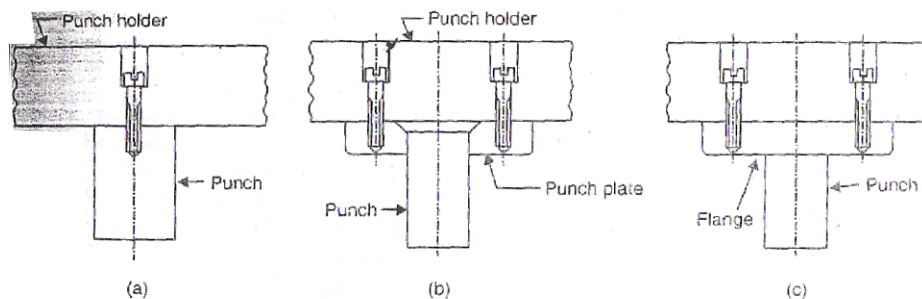


Fig. 2.33. Methods of Holding Punches.

Strippers

After a blank has been cut by the punch on its downwards stroke the scrap strip has the tendency to expand. On the return stroke the scrap has the tendency to adhere to the punch and lifted by it. This action interferes with the feeding of the stock through the die and some device must be used to strip the scrap material from the punch as it clears up the die block. Such a device is called 'stripper' or 'stripper plate' striping loaded or movable, (fig. 2.35). As is clear, it is plate parallel with and above the die surface. An opening is cut through the stripper plate for free

passage of the punch. By rule of thumb, this opening can be about 1.6 mm large than the blank size on all sides.

- a. **Fixed stripper [Fig. 2.35 (a)].** This stripper is attached at a fixed height over the die block. The height should be sufficient to permit the sheet metal to be fed freely between the upper die surface and the under surface of the stripper plate the stripper plate is usually of same width and length as the die block. In simple dies, it is fastened with the same screws and dowels which are used for die block. In complex dies, the stripper fastening will be independent of die fastening. The thickness of the stripper plate should be sufficient to withstand the force needed to strip the scrap strip from the punch. The usual value is 9.5mm to 16mm.

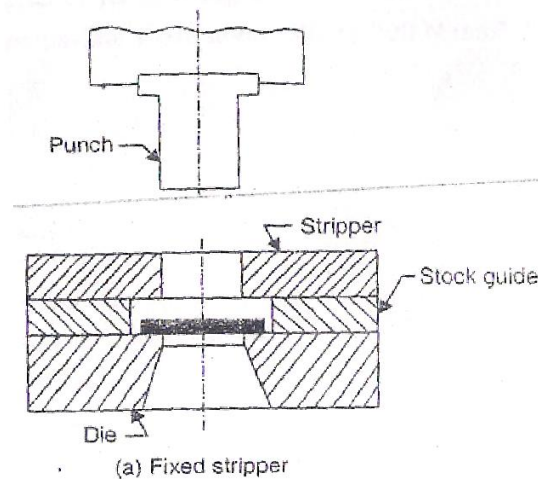


Fig. 2.35. Strippers

The following empirical formula may also be for determining stripper plate thickness

$$t_{\text{str}} = \frac{1}{8}(w/3 + 16t)$$

Where w and t are the width and thickness of stock strip.

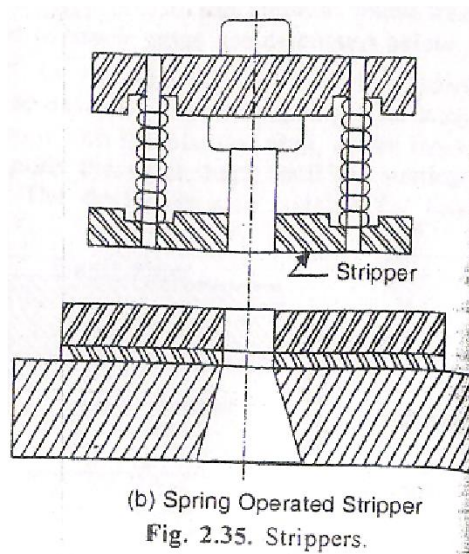
$$\begin{aligned} \text{Or } t_{\text{str}} &= 0.5 \times T, \text{ if guided} \\ &= 0.75 \times T, \text{ if un-guided} \end{aligned}$$

The thickness of the stripper plate may also be determined by the size of the socket-head can screw used to hold it in place. The stripper plate must be thick enough to allow for the screws, dowels are used to ensure accurate alignment of the die block. Stripper may be made of cold rolled M.S. The fixed stripper is also known as channel stripper.

For the stripping action upward movement of the punch, the scrap will strike the underside of the stripper plate and get stripped off from the punch. The underside of the strip plate which comes in contact with the strip should be machined and preferably ground. The height of the stock strip channel should be at least equal to 1.5 times the stock thickness. If the scrap strip is to be lifted over a fixed pin stop, this height should be increased. The width of the channel should be equal to the width of the stock plus adequate clearance. The disadvantages of fixed stripper are that it hides the work from the operator and it would interfere with removal of the scrap strip in large blanking operations.

- b. **Spring loaded stripper.** [Fig. 2.35 (b)] this type is used on large blanking operation and on highly ductile materials where it is desirable to utilize the pressure to hold the surrounding stock during the blanking operation. In this design the stripper plate is mounted over compression springs and suspended by bolts from the punch holder, with the lower surface of the plate of the stripper below the cutting end of the punch. As the punch travels

downward for the blanking operation, the stripper plate contracts the stock strip first and holds it until the punch clear the strip on its return stroke. As the punch rise spring pressure holds the strip, stripping it from 2.5 to 20% of cutting force. However, the more common value for most of the applications is 5 to 10%.



The stripping force may also be determined with the help of the following relation.

$$F_{str} = spt, \text{ KN}$$

Where p and t are in mm and s is the stripping constant, given below:

$S=0.015$ for low C-steels with sheet metal under 1.6 mm thick Where the cut is close to the edge of the stripe or preceding cut.

$=0.0145$ for the same materials but for other cuts

$=0.021$ for the same material with $t > 1.6$ mm

$=0.24$ for harder materials.

$$F_{str} = (0.05 \text{ to } 0.08) \times F \text{ for normal clearance}$$

$$=(0.10 \text{ to } 0.20) \times F, \text{ for narrow clearance and complicated contours}$$

Comparison of fixed and spring loaded strippers:

1. Solid stripper is simple in construction and is cheap.
2. Fixed or solid stripper does not apply any hold-down pressure like a spring stripper. Due to this thinner sheets may buckle and jam in the stripper. So it is mainly used for thick sheets
3. Clearances in the fixed stripper channel allow the sheet to shift slightly. Additional edge stock is necessary to insure complete blanks being cut.
4. Fixed strip is used mainly for strip sheet metal where feeding of the sheet metal accomplished by hand. Coiled sheet metal uses a feed roll and spring stripper in most cases.
5. For every large stripping force, fixed stripper is preferred because it may not be possible to provide enough stripping force with springs in the space available.
6. The travel of spring strippers is limited by the compression allowance of the springs
7. In spring strippers. The full stripping force is not available throughout the stripping action it is build up as the springs are compressed.
8. Additional die space is required for placing the strings around the punch.

Stock stop

The strip of sheet metal is fed and guided through a slot in the stock guide fig. 2.35 (a) or through a slot in the stripper plate. After each blanking, the strip has to be advanced a correct distance. The device used to achieve this is called stock stop. The simplest arrangement may be a dowel pin or a small block, against which an edge of the previously blanked holed, is pushed after each stroke of the press. On its upward stroke, the punch carries the stock strip as far as the underside of the stripper plate. Due to this, the stock strip gets released from the stop. With constant pressure exerted pushing the stock strip to the left, the stock will move as it lifted clear, then drop with the next hole over the hole over the stop as the scrap strip is stripped from the punch, Fig. 2.36.

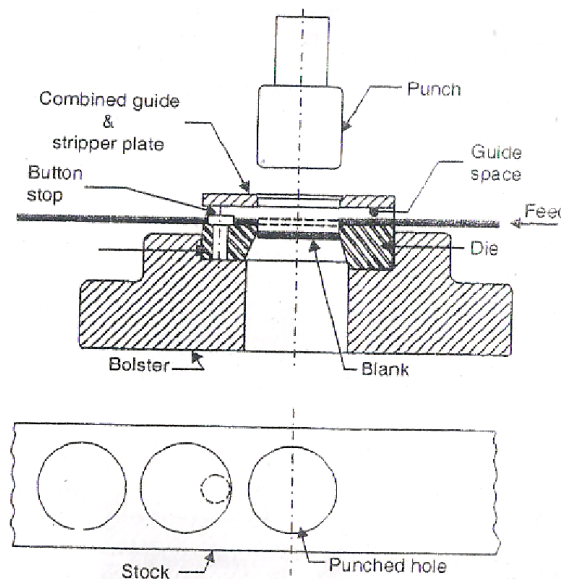


Fig. 2.36. Stock Stop.

This type of stock stop is suitable for only low and medium production dies and not for high production dies, since the operator has to force the stock over the shoulder to secure a desired feed length. Also this

needs considerable skill on the part of the operation. This stop is also not suitable where frail die sections would be damaged by a misfeed. Other types of stock stops are discussed below.

1. Latch stop or Trip stop or pawl stop. Fig. 2.37: In these types of stop, a latch or pawl is pivoted on a pin fitted into a block on stripper and is held down by a tension spring. The latch is lifted by the scrap bridge on the ratchet principle and drops into the blanked area, as the stock is fed forward manually into the die. The operator then pulls the stock back until the vertical surface of latch bears exactly against the scrap bridge. The design is also suitable for low production only.

2. Automatic stop: In this design also, a hinged latch or lever (finger) is used. The working of an automatic stop is shown in Fig. 2.38. An adjustable strip guide is fastened to the punch holder. During the

downward stroke of the slide, as the punch cuts the blank, this strip screw strikes the end of the latch, thereby lifting its pin end. On the return stroke, the pin end drops. The finger pivoted loosely, so that it can have endwise movement. Due to this endwise movement of the finger, its pin end does not drop into its former position, but drops on to the top surface of the stock. As the stroke is fed forward, the pin drops into the next blank space. This enables the operator to locate the stock strip exactly. Another design of automatic stop is show in Fig. 2.39.

In this design, as the gauge pin is raised from the blank space due to lever action, it rocks so, as the stock is fed forward, the gauge pin rests on the top surface of the stock until it drops into the next blanked hole.

3. **Solid stops shoulder stops:** Solid stops are extensively used on progressive dies when the operation is a cut off or trimming one to position the end of the stock or the work piece. Two designs are used in Fig. 2.40. In Fig. (a), the stop is fastened to the die, where as in Fig. (b) it is fastened to the stripper.

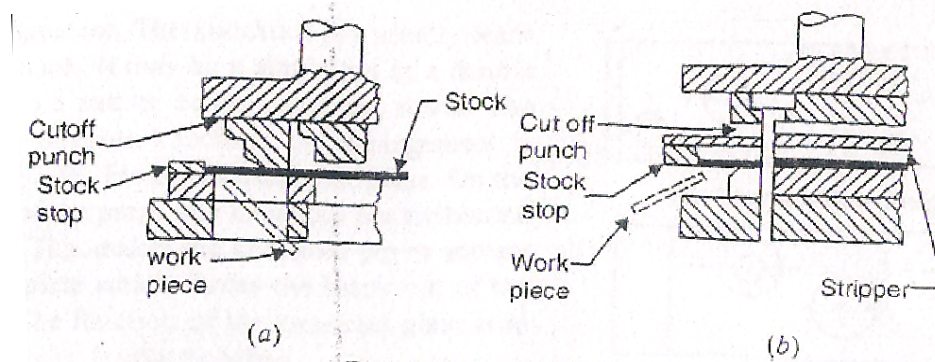


Fig. 2.40. Solid Stops.

4. **Starting stops:** A starting stop or a primary stop is used to position stock as it is initially fed into the die. It is mounted on the stripper plate, Fig. 2.41. It consists of a latch, which is pushed inward by the operator by one end until it contacts the stripper plate. The stock strip which is being fed by the operator with his other hand is made to locate against the latch. The latch is held in position until the first die operation is completed and then the latch is released.

Note. Stops are needed when the stock fed manually, in the case of automatic feeding, die stops are usually not needed as the strip can be advanced quite accurately.

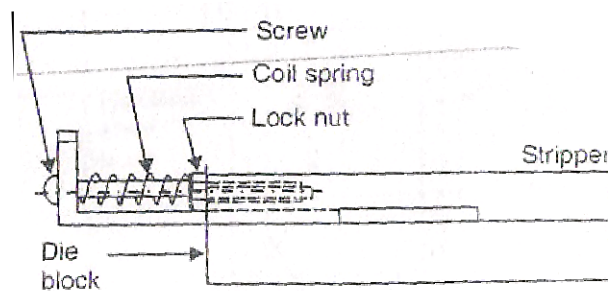


Fig. 2.41. Starting Stops.

Stock Guide

Stock guide is the space provided in the die-block through which the stock strip is guided as it is fed in to the die. The design of the stock guide will depend upon the type of stripper. For fixed strippers (channel strippers), the channel for guiding the stock is shown in fig. [2.35(a) and 2.36] and its design is given under “fixed stripper operated strippers, the various designs for guiding the stock are shown in fig. 2.42 (a, b, c). in the figure. Guide rails are mounted on the die block for guiding the stock. In fig. 242 (a), the stripper acts as a pressure pad and presses directly against the stock strip. The stripper does not contact the guide rails. In fig. 2.42 (c), hooks are provided on the guide rails in order to improve the guiding effect when there is space limitations but stock-strip guides may be used fig. 2.42 (d). At least three button guides should be used on each side of the stock strip. The guiding not as efficient as with guide rails.

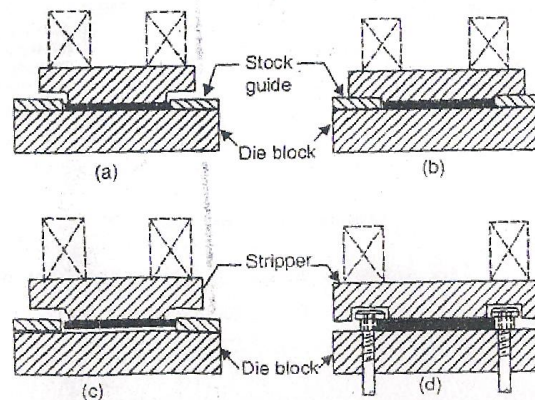


Fig. 2.42. Stock Guides with Spring Operated Strippers.

Strip Feeding

Stock scrip may be fed into the die either manually or mechanical at (automatic feeding) manual feeding is suitable only for low production or

with presses operating at low values of strokes per min. modern presses operate at 200 to 300 strokes per min. (maybe up to 500 strokes per min.) For such cases, manual feeding is not feasible and automatic feeding is the only answer. For this, the strip is prepared in large coils. The first step in the feeding of strip is the unwinding of the coil. Two methods are in use for this purpose: (a) Reel (b) coil cradle.

The reel is considered to be better as it does not damage the strip any way. The reel may be or may not be power driven. In the case of power driven. In the case of power driven reels a roller at the end of a loop arm, rides on the uncoiling strip. When sufficient coil has been unwound, the loop and actuates a switch which stops the power drive. Now as the strip is used up and the loop arm get raised, the power supply is switched on. In the case of unpowered reels, the coil is unwound by an external power source, which may be feeding mechanism or straightening rolls. When enough coil has been unwound, the loop arm get raised, the power supply is switched on. In the case of unpowered reels. The coils, when enough coil has been unwound the reel is stopped from uncoiling by a manual or automatic brake.

In the case of cradle, the strip is supported on the outside diameter of the coil. The coil locates against rollers and due to this scratches may form on the coil.

The second step in stock feeding is straightening the coil; it is passed through between a series of rollers.

After uncoiling and straightening, the final step is to feed the strip into the die. The two main types of feeding system are:

1. Roll feed
2. Slide or hitch feed

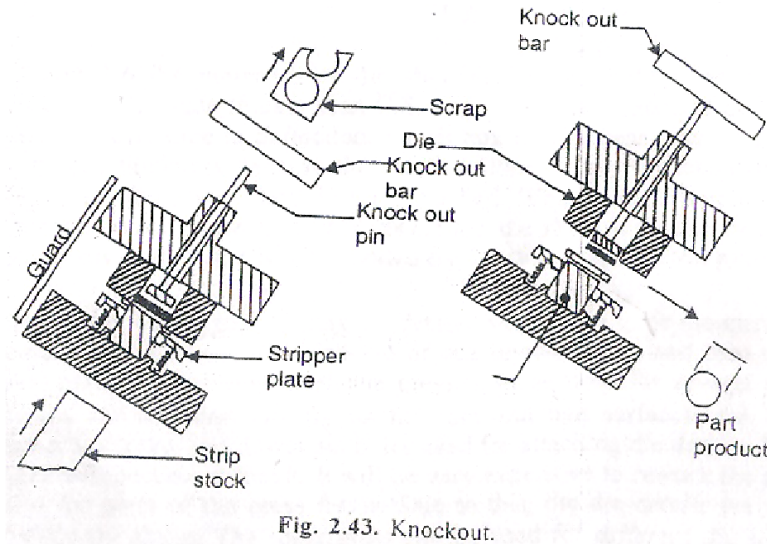
The roll feed moves the stock between a pair of rollers, which is driven through overrunning clutch or ratchet mechanism timed from the press main shaft or ram. Roll feed may be: single or single or double. In the case of single roll rollers are provided only on one side of the press and they push or pull the strip through the die. In the case of double roll feed, feeding mechanism is mounted on each side of press bed with a drive connection between them. One feed pushes the stock and the other pulls stock through the die. This will keep strip and prevent its buckling.

In the case of slide or latch feed, grippers are used which grasp the strip mechanically and feed it into the die by a reciprocating mechanism which may be driven from the press crankshaft a cam mounted on the punch holder, hydraulically or pneumatically

Knockouts

As already stated, the function of a knockout is to shed or eject work piece from within the die cavity as the work piece may get jammed in the die cavity due friction. A knockout may be actuated by springs (Fig. 2.25) or by a positive and bar arrangement. The knockout pin usually leads through the shank. It may be a single pin or double pin fastened to a pad

or collar above the shank. The working Fig. 2.43, for an inverted OBI press. On the return stroke of the press, the knockout pin press knockout pin strikes the knockout plate bar. This makes the knockout pin to actuate the knockout plate which force the blank out of the die opening. The function of the knockout plate is to support and guide fragile punches.



Die-set

The die shoe, the punch holder together with two or more guide post constitutes a die set Fig. 2.44. The die shoe and the punch holder are made of C.I.C.S and rolled steel. For smaller dies, CI is used, whereas for larger and special die sets, C.S and rolled toll steal are used. Bushings are assembled to the upper shoe (punch holder) by press fitting and guide posts are press fitted into the lower shoe (die shoe). The bushings and posts are sized to provide a slip fit. For average range of die sets, the

diameter of guide posts varies from 2.5 cm to 7.5 cm larger pins may be used if extreme alignment is required. When die is full closed, the upper ends of the guide post should not project beyond the top surface of the upper shoe.

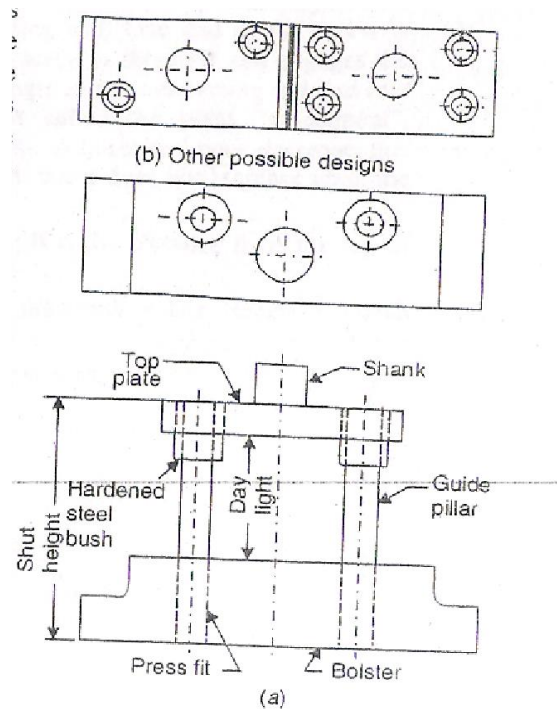


Fig. 2.44. Die Set.

A typical bottom bolster (die shoe) plan is shown in fig. 2.45

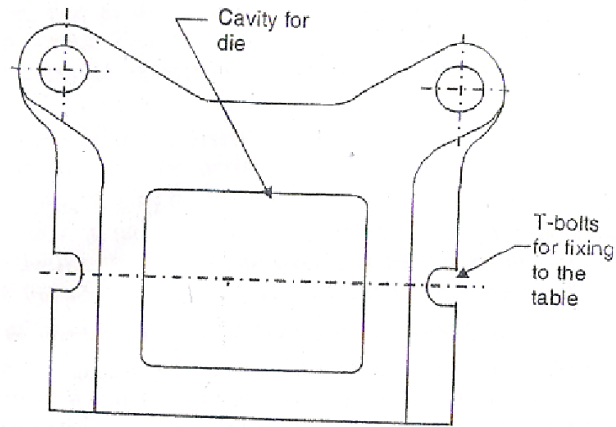


Fig. 2.45. Bottom Bolster.

2.20 SOLVED EXAMPLES

Example 2.1.

Find the total pressure, dimensions of tools to produce a washer 5 cm outside diameter with a 2.4 cm diameter hole, from material 4 mm thick having a shear strength of 360 N/mm²

Solution.

The production of washer consists of two operations. Blanking the outside diameter and piercing the inner hole.

Blanking pressure.

Given,

$$\pi = \frac{22}{7} = 3.143$$

$$D = 5\text{cm} = 50\text{mm}$$

$$t = 4\text{mm}$$

$$\tau_s = 360\text{N/mm}^2$$

$$f_1 = \pi D t \tau_s$$

$$= \pi \times 50 \times 4 \times 360$$

$$= 226285.71429 \text{ N/1000}$$

$$f_1 = 226 \text{ KN}$$

Given,

$$\pi = \frac{22}{7} = 3.143$$

$$d = 2.4\text{cm} \times 10 = 24 \text{ mm}$$

$$t = 4\text{mm}$$

$$\tau_s = 360 \text{ N/mm}^2$$

$$F_2 = \pi d \tau_s$$

$$= \pi \times 24 \times 4 \times 360$$

$$= 108617.14286 \text{ N/1000}$$

$$= 108.768 \text{ KN}$$

$$F = F_1 + F_2$$

$$= 226.6 + 108.7$$

$$335.4 \text{ KN}$$

$$= 335.4 \text{ KN}$$

Clearance is taken as 10% of stock thickness. Therefore

$$\text{Piercing punch diameter} = 2.4 \text{ cm}$$

$$\text{Piercing die diameter} = 2.4 + 0.4 = 2.8 \text{ cm}$$

$$\text{Blanking die diameter} = 5 \text{ cm}$$

$$\text{Blanking punch diameter} = 5 - 0.4 = 4.6 \text{ cm}$$

The above calculations give the press capacity for tools without shear.

When suitable shear applied the loads calculated can be reduced by 20%.

$$\text{Press capacity} = 0.8 \times 335.4 = 268.32 \text{ KN}$$

The blanking punch will be flat and shear is applied, the loads calculated can be reduced by 20%

$$\text{Press capacity} = 0.8 \times 335.4 = 268.32 \text{ KN}$$

$$\frac{20}{100} \times 335.4 = 68.08 \text{ or } 335.4 - 268.32 = 68.08 \text{ KN}$$

Example 2.2.

A cup without flanges and of height 10 cm and diameter 5 cm is to be made from sheet metal 2.5 mm thick find the suitable number of draws.

Solution.

First of all, blank diameter should be obtained.. it is given as:

$$D = \sqrt{a^2 + 4dh}$$

$$D = 5 \text{ cm and } h = 10 \text{ cm}$$

$$D = 15 \text{ cm}$$

The height diameter ration of cup is 2. Therefore, from table 2.9, the tentative number of draws 3. Let the reduction be: 45%, 25% respectively.

$$1^{\text{st}} \text{ reduction} = 45\% = 15 \times 0.45 = 6.75$$

$$\text{Diameter } d_1 \text{ at first draw} = 15 - 6.75 = 8.25 \text{ cm}$$

$$2^{\text{nd}} \text{ reduction} = 25\% = 8.25 \times 0.25 = 2.06 \text{ cm}$$

$$\text{Diameter } d_2 \text{ at second draw} = 8.25 - 2.06 = 6.19 \text{ cm}$$

$$3^{\text{rd}} \text{ reduction} = 20\% = 6.19 \times 0.2 = 1.238 \text{ cm}$$

$$\text{Diameter } d \text{ at final draw} = 6.19 - 1.238 = 4.952 \text{ cm}$$

To get exactly the value of 5 cm, the third reduction can be slightly lower 20% (19.25 percent exactly).

Example 2.3.

A 37.5 cm long 19 mm wide and 2.5 mm thick strip is to be bent in a v- shaped die. Calculate the bending force necessary if the steel has 630 N/mm, tensile strength.

Solution.

Let the width of die opening be $16t$, I.e,

$$W = 16 \times 2.5 = 40 \text{ mm}$$

Now bending force is given as,

$$F = \frac{k.I\sigma_{,,}t^2}{w}$$

Now $k = 1.20$, $I = 37.5 \text{ cm}$, $t = 2.5 \text{ cm}$

An $\sigma_{,,} 6030 \text{ N/mm}^2$

$$F = \frac{1.2 \times 35 \times 10 \times 630 \times (2.5)^2}{40}$$

$$44.3 \text{ KN}$$

Example 2.4 estimate the blanking force to cut a blank 25 mm wide and 30 mm long from a 1.5 mm thick metal strip, if the ultimate shear stress of the material is 450 N/mm². Also determine the work done if the percentage penetration is 25 of material thickness.

Solution.

$$(a) \text{Perimeter of black} = 2(25 + 30)$$

$$= 110 \text{ mm}$$

$$\text{Blanking force} = \text{Area under shear} \times \text{ultimate shear stress}$$

$$= \text{perimeter} \times t \times \tau_s$$

$$= 110 \times 1.5 \times 450$$

$$=74.25 \text{ KN}$$

(b) Punch travel = percentage penetration

$$= \frac{1}{4} \times t = \frac{1.5}{4} = 0.375 \text{ mm}$$

Work done = force $\times$ punch travel

$$= 74.25 \times 10^3 \times 0.375 \times \frac{1}{10}$$

$$= 27.84 \text{ NM}$$

Example 2.5

a washer with a 12.7 mm internal hole and an outside diameter of 25.4 mm is to be made from 1.5 mm thick strip of 0.2 per carbon steel. Considering the elastic recovery of the material find: (a) the clearance (b) blanking die –operation size (c) the blanking punch size (d) the clearance (one side) for soft steel.

Solution.

(a) The clearance (one side) for soft steel

$$= 5\% \text{ of } t$$

$$C = 0.05 \times 1.5$$

$$= 0.075 \text{ mm.}$$

(b) It is clear from fig. 2.20 (b), the blanking die opening size is equal to the blank size but to allow for the expansion of the blank, the die opening should be made smaller. Thus,

$$\text{Blanking die opening} = 25.40 - 0.05$$

$$= 25.35 \text{ mm}$$

(c) Blanking punch size = blanking die opening - 2c

$$= 25.35 - 0.15$$

$$= 25.20 \text{ mm}$$

- (d) The punch size determines the size of the hole produce. To allow for the contraction of the hole due to elastic recovery, the punch size should be made larger by an amount 0.05 mm (see article 2.8.1) so the piercing punch size will be,

$$\text{Piercing punch size} = 12.70 + 0.05$$

$$= 12.75 \text{ mm}$$

- (e) Piercing die size = piercing punch size + 2c, [fig. 2.22 (a)]

$$= 12.75 + 0.15$$

$$= 12.90 \text{ mm}$$

Example 2.6.

the ultimate shearing of the material of the washing in example 2.5 is 280 N/mm^2 (a) find the total cutting force if both punches act at the same time and no shear is applied to either punch or the die. (b) what will be the cutting force punches are of 1 mm , what will be the cutting force if both punches act together.

Solution.

$$(a) F = \pi (D + d) t \tau,$$

$$D = 25.4 \text{ mm}, d = 12.7 \text{ mm } t = 1.5 \text{ mm}$$

$$\tau = 280 \text{ N/mm}^2$$

$$F = \pi (25.4 + 12.7) \times 1.5 \times 280$$

$$= 50.27 \text{ kN}$$

- (b) When the punches are staggered, the punch taking the large cut will require the greatest force.

$$\begin{aligned}
 F &= \pi D t \tau_s \\
 &= \pi \times 25.4 \times 1.5 \times 280 \\
 &33.5 \text{ KN}
 \end{aligned}$$

$$F = \frac{t \times k \text{ rxf mm}}{(k \times t + I)}$$

$$K = \text{percentage of penetration} = 0.60$$

$$I = \text{shear on punch} = 1 \text{ mm}$$

$$F_{\max} = 50.2 \text{ kN}$$

$$F = \frac{15 \times 0.6 \times 50.27}{(0.9 + 1)}$$

$$= 23.8 \text{ kN}$$

Example 2.7.

A hole of 60 mm diameter is to be produced in steel plate 2.5 mm thick. The ultimate shear strength of the plate material is 4050 N/mm². If the punching force is to be reduced to half of the force using a punch without shear, estimate the amount of shear on the punch. Take percentage penetration as 40%.

Solution.

The punching force with non-sheared punch.

$$\begin{aligned}
 F_{\max} &= \pi D t \tau, \\
 &= \pi \times 60 \times 2.5 \times 450 \\
 &= 212 \text{ KN}
 \end{aligned}$$

$$\text{Work done} = f_{\max} \times \text{penetration (punch travel)}$$

$$F_{\max} = \pi D t \tau,$$

$$= 212 \times 10^3 \times 0.4 \times 2.5 \times \frac{1}{10}$$

$$= 212 \text{ Nm}$$

Now work done remains the same with a sheared and a non-sheared punch. For a sheared punch.

$$\text{Punch travel} = \text{penetration} + \text{shear}$$

$$= K, t + I$$

If F is the blanking force, then comparing the work done.

$$F_{\max} \times K \times t = F (K.t + I)$$

$$I = \frac{K \times t (F_{\max} - F)}{F}$$

Now $F = \frac{1}{2} F_{\max} = 106 \text{ KN}$

$$I = \frac{0.4 \times 25 \times 106}{106}$$

$$= 1 \text{ mm}$$

A mount of shear on punch = 1 mm.

Example 2.8.

in fig. 2.27, the strip thickness is 3.2 mm and the dimension 'h' of the blanking is 10 cm and is 2.5 cm find.

- i. The value for back scrap.
- ii. The value for scrap bridge.
- iii. The width of strip.

- iv. The length of one piece of stock needed to produce on part
- v. The number of parts which can be produced from a 2.4 m long strip
- vi. The scrap remaining at the end of the strip.

Solution.

- (i) Back scrap and the front scrap are given as,

bridge and the

$$a = t + 0.015 h$$

Here $t = 3.2 \text{ mm}$ and $h = 10 \text{ cm}$

$$a = 3.2 + 0.015 + 100 = 4.7 \text{ mm}$$

- (ii) The scrap bridge and the distance between successive blanks given as, see. Table 2.1

$$b = t = 3.2 \text{ mm}$$

- (iii) $w = h + 2a$

$$a = 3.2 + 0.94 = 10.94 \text{ cm say } 11.00 \text{ cm}$$

- (iv) The length of one piece of stock needed to produce one blank is,

$$\begin{aligned} S &= w h + b \\ &= 2.5 + 0.32 = 2.82 \text{ cm} \end{aligned}$$

- (v)
$$N = \frac{L-b}{s}$$
- $$= \frac{240-0.32}{2.82} = 84 \text{ blanks}$$

- (vi) Scrap remaining at the end,

$$y = L - (Ns + b)$$

$$y = 240 - (82 \times 2.82 \times 0.32)$$

$$= 240 - 237.2 = 2.80 \text{ mm}$$

Example. 2.9.

The symmetrical-cup work pieces shown in fig. 2.3, is to be made from cold rolled steel 0.8 mm make the necessary calculation for designing the drawing die for this part.

Solution.

The shell diameter $d = 50$ mm radius of bottom corner, $r = 1.6$ mm

- i. Size of blank: the ration.

$$\frac{d}{r} = \frac{50}{1.6} = 31.25$$

So, the formula for determining the shell blank size will be Eqn 2.16

$$D = \sqrt{d^2 + 4dh}$$

$$D = 50 \text{ mm}, h = 50 \text{ mm}$$

$$D = 111.80 \text{ mm}$$

This is the theoretical blank size, however, to get a smooth edge, a small trimming is necessary. For this, it is necessary to add extra metal. Rule of thumb is to add about 3.2 mm to the blank diameter for each 2.5 cm of cup diameter is 50 mm, 6.4 mm should be added for trimming.

$$D = 111.80 + 6.40 = 118.2 = 118.5 \text{ mm}$$

- ii. Percentage reduction. Percentage reduction is given as,

$$= 100 \left(1 - \frac{d}{D} \right)$$

$$= 100 \left(1 - \frac{50}{118.5} \right) = 57\%$$

As per practice, a reduction of about 45 to 50 per cent is permission for first draw. Therefore, it is clear that the above cup cannot be drawn in one draw.

iii. Number of draw:

$$\text{Height to diameter ration} = \frac{50}{50} = 1.0$$

Therefore, from Table 2.9,

$$\text{Number of draws} = 2$$

Let the first reduction be 45%

$$\text{Diameter } d, \text{ at first draw} = 1118.5 - 0.45 \times 118.5 = 65.175 \text{ mm}$$

Diameter at the end of second draw = has to be 50 mm

Therefore, reduction for second draw.

$$\begin{aligned} &= 100 \left(1 - \frac{50}{65.175}\right) \\ &= 23.28 \% \end{aligned}$$

Which is less than 30% the permissible reduction for the draw.

iv. Radius on punch and die: for the first draw, the punch radius at the bottom should be at least $= 4 \times t$

$$= 4 \times 0.8 = 3.2 \text{ mm say } 4 \text{ mm}$$

For the second draw, the punch radius, is determined by the corner radius of the finished shell, that is it, should be equal to 1.6 mm.

The die radius, r (fig. 2.54) should be 4 to 10 times the stock thickness.

$$r, = 3.2 \text{ mm to } 8 \text{ mm, say } 6 \text{ mm}$$

- v. Die clearance: From Table 2.7, die-clearance is given as

For the first draw = $1.08 t$ to $1.1 t$ per side

$$= 0.864 \text{ to } 0.88 \text{ mm say } 0.87 \text{ mm}$$

For the second draw = $1.09 t$ to $1.12 t$

$$= 0.872 \text{ to } 0.896 \text{ mm, say } 0.88 \text{ mm}$$

Punch diameter for the second draw

$$= 65.175 - 2t$$

$$= 65.575 \text{ mm}$$

Die opening diameter for the second draw

$$= 65.575 + 2 \times 0.87$$

$$= 65.315 \text{ mm}$$

- vi. Drawing pressure: $f = \pi dt \cdot \sigma_y \left(\frac{D}{d} - c \right)$

$$D = 50 \text{ mm}$$

$$t = 0.8 \text{ mm}$$

$$c = \text{constant (0.6 to 0.7)}$$

$$\sigma_{yt} = 427 \text{ N/mm}^2$$

$$F = \pi \times 50 \times 0.8 \times 427 (2.37 - 0.65)$$

$$= 92.30 \text{ kN}$$

- vii. Blank holding pressure:

Blanking holding pressure may be taken as one-third of drawing pressure, i.e, 30.8 kN. Blank holding pressure can be found by the relation, for cylindrical draws,

$$F_{bh} = \frac{\pi}{4} (D^2 - d^2) \cdot p_{bh} \text{ where}$$

p_{bh} is the specific blank holding pressure and can be taken as (N/mm²),

for M.S, ($t < 0.5$ mm) = 2.5 – 30; M.S. ($t > 0.5$ mm) = 2.0 -2.5

brass = 1.5 -2.0; cu =1.0 – 2.0; Al =0.8 – 12

viii. Press capacity:

A 150 kN capacity press will be alright for the total possible maximum pressure of about 12 kN

Example. 2.10.

determine the developed length of the part shown in fig. 2.74.

Solution.

While calculating the development length for bending external dimensions should be converted to internal dimensions, fig 2.4 (b)

$$\text{Developed length} = L_1 + L_2 + B$$

The inside radius of bend, $r = 3.2 - 2.3 = 0.90$ mm

$$\text{Length } L_1 = 76 - (2.3 + 0.90) = 72.8 \text{ mm}$$

$$\text{Length } L_2 = 115 - (2.3 + 0.90) = 111.8 \text{ mm}$$

$$\text{Bend allowance, } B = \frac{a}{360} 2\pi (r + k)$$

$$\text{Here } a = 90^\circ$$

$$\text{Let } K = t/3$$

$$B = \frac{90}{360} 2\pi (0.90 + \frac{2.3}{3})$$

$$2.61 \text{ mm}$$

$$\text{Development length} = 72.8 + 111.8 + 2.61$$

$$= 187.21 \text{ mm}$$

Example 2.11.

calculation the bending force for channel bending for the following data thickness of blank = 3.2 mm

Bending length = 900 mm

Die radius = punch radius = 9.5 mm

Ultimate tensile strength of the material = 400 N/mm²

Solution.

Bending force is given as:

$$F = \frac{k.l.\sigma_{ul} t^2}{w}, N \quad (\text{Eqn. 2.23}).$$

Now K = 0.67 for channel bending

$$l = 900 \text{ mm}$$

$$\sigma_{ul} = 400 \text{ N/mm}^2$$

$$t = 3.2 \text{ mm}$$

$$w = R_1 + R_2 + C \quad [\text{fig. 2.65 (b)}]$$

$$= 9.5 + 9.5 + 3.2 = 22.2 \text{ mm}$$

$$F = \frac{0.67 \times 400 \times 3.2 \times 3.2}{22.2}$$

$$= 111.25 \text{ kN}$$

Example 2.12.

calculation the bending for a 45° bend in aluminum blank. The following data is given.

$$\text{Blank thickness} = 1.6 \text{ mm}$$

Bend length = 1200 mm

Die –opening = 8 × metal thickness

Ultimate tensile strength = 455 N/mm²

Solution.

Bending force is given as:

$$F = \frac{k.I.\sigma_{ut} t^2}{w} \text{ N}$$

$$K = 1.33$$

$$I = 1200 \text{ mm}$$

$$\sigma_{ut} = 455 \text{ N/mm}^2$$

$$t = 1.6 \text{ mm}$$

$$f = \frac{1.33 \times 1200 \times 1.6 \times 1.6}{12.8} \text{ N}$$

$$= 145.24 \text{ kN}$$

Example 2.13.

determine the capacity of the double bending for the following data sheet

metal thickness (cold rolled steel) = 1 mm

Sheet metal width at bend = 50 mm

Die radius = 3 mm

Punch radius = 1.5 mm

Die clearance = 1.25 mm

Tensile pressure 315 MP_a

Bends on punch = 2

Project width of each bead = 3 mm

Solution.

i. Bending force: $F = \frac{(0.67 l \sigma_{ut} t^2)}{w}$, $w = R_1 + C + R_2$

$$= \frac{0.67 \times 50 \times 1 \times 1}{3 + 1.25 + 15 + 15} = 18352, \text{ N}$$

ii. Pad force $= 0.67 \cdot \sigma_{ut} l \cdot t$

$$= 0.67 \times 315 \times 50 \times 1$$

$$= 10552.5, \text{ N}$$

iii. Bottoming $= \sigma_c l \cdot b$.

$$= 560 \times 50 \times 2 \times 3$$

$$= 168,000 \text{ N.}$$

It is clear from above, that the actual bending force is minor as compared to other force. Now, bending force occurs above bottom of press stroke. Pad force occurs through bending operating and bottoming force occurs at bottom of press stroke. Mechanical press rated at the bottom of the stroke. So, in view of its magnitude, when bottom is used alone to determine the press capacity.

$$\text{Force required for the operation} = 10552.5 + 168,000$$

$$= 178552.5, \text{ N}$$

$$= 18.2 \text{ tonnes}$$

Used a 20tonne press.

If bottoming had not been used, then force required would be,

$$\text{Force required} = 1835.2 + 10552.5$$

$$= 12387.8, \text{ N}$$

$$= 1.263 \text{ Tonnes.}$$

Example 2.14.

A.M.S sheet 5 mm thick and 2 m wide is cut in the width direct determine cutting force force for cutting:

- (a) With paraller cutting edges.
- (b) In a guillotine in wich cutting edge are given a shear of 6° take ultimate stress in shear as 382.5 MP_a .

Solution.

- (a) Area to be cut = width $\times$ sheet thickness

$$\begin{aligned}\text{Cutting force} &= 2 \times 1000 \times 5 \times 382.5, \text{ N} \\ &= 3.825 \text{ MN.}\end{aligned}$$

- (b) In this case, the length to be cut is the instantaneous length to be cut, which is

$$\begin{aligned}&= \frac{t}{\sin a} = \frac{5}{\sin 6} \\ &= 47.83 \text{ mm}\end{aligned}$$

$$\begin{aligned}F &= 4.83 \times 5 \times 382.5, \text{ N} \\ &91.48 \text{ kN}\end{aligned}$$

It should b note that there is a large drop in the cutting force.

Example 2.15

A cup 105 mm inside diameter and 90 mm deep is to be drawn from steel sheet 1mm hick. Determine the blank diameter and suitable punch diameter for the first draw.

Given the probable dimension of the cup obtained from the draw and estimate the press capacity.

Solution.

Blank diameter,

$$D = \sqrt{d^2 + 4dh} = [105^2 + 4 \times 105 \times 90]^{1/2} = 221 \text{ mm}$$

$$\text{Thickness ratio, } \frac{t}{D} \times 100 = \frac{1}{221} \times 100 = 0.453$$

From the table below:

$$\text{Thickness ratio } \frac{t}{D} \times 100: 0.15 \quad 0.20 \quad 0.30 \quad 0.40 \quad 0.50 > 0.50$$

$$\text{Max draw ratio, } D/d: 1.43 \quad 1.54 \quad 1.67 \quad 1.82 \quad 1.91 \quad 2.00$$

(for sheet steel where pressure pad is used)

For thickness ratio of 0.453, the safe drawing ratio is 1.82

$$\text{Diameter for the first draw} = \frac{221}{1.82} = 121 \text{ mm.}$$

The cut diameter is 150 mm, hence it can not be drawn by a single draw.

If d' is taken as 121 mm for the first draw, the reduction for the subsequent draw will be rather small. So it will be better to have a somewhat lower reduction for the first draw.

Let the diameter of the first draw be $d = 130 \text{ mm}$

$$\frac{D}{d} \text{ for first draw is } = \frac{221}{130} = 1.7$$

$$\frac{D}{d} \text{ for second draw } = \frac{130}{105} = 1.24, \text{ which is quite O.K.}$$

$$\text{Depth of first draw: } 130^2 \times (4 \times 130) \times h = 221^2$$

$$\text{From here } h = 61.43 \text{ mm}$$

$$\text{Press capacity} = \pi d t \sigma \left(\frac{D}{d} - C \right) = \tau \times 130 \times 1 \times 415 \left(\frac{221}{130} - 0.65 \right) =$$

$$17.45 \text{ kN.}$$

Example 2.16.

A symmetrical cup of 8 mm diameter and 250 mm height is to be fabricated on a deep drawing die. How many drawing operations will be necessary if no intervention annealing is done. Also find the drawing force.

Solution:

Blanking diameter,

$$D = \sqrt{d^2 + 4dh}$$

$$= \sqrt{64 + 4 \times 8 \times 25} = 29.4 \text{ cm}$$

Let $\frac{D - D_1}{D} = 0.5$ for the first draw

Diameter after the first draw,

$$D_1 = D - 0.5 D = 0.5 D = 14.7 \text{ cm}$$

Let the reduction for the second draw be 40%.

Diameter after the second draw

$$= 14.7 - 0.4 \times 14.7 = 8.82 \text{ cm}$$

Now, final diameter = 8 cm.

Third draw is also necessary

Percentage reduction for the 3rd draw

$$= 8.82 - R \times 8.82 = 8$$

$$1 - R = \frac{8}{8.82} = 0.9$$

$$R = 10\%$$

Now, $\pi D_1 l_1 = \frac{\pi}{4} \times (14.7)^2$

$$= 509 \text{ cm}^2$$

Height of cup after draw

$$= \frac{509}{\pi \times 14.7} = 11.02 \text{ cm}$$

Similarly $l_2 = 22.28 \text{ cm}$

And $l_3 = 25 \text{ cm}$

Now drawing force $F = \pi \cdot d \cdot t \cdot \sigma_{yt} \left(\frac{D}{d} - C \right)$

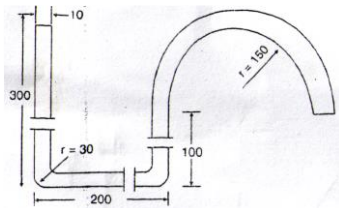
Let $t = 3 \text{ mm}$, $\sigma_{yt} = 250 \text{ N/mm}^2$ and $C = 0.66$.

$$F = \pi \times 80 \times 250 \left(\frac{294}{80} - 0.66 \right)$$

$$= 56.556 \text{ kN}$$

Example 2.17

determine the deployment length of the part shown in fig. 2.27 (q)



Solution:

While calculating the deployment for being, the external dimension converted into internal dimension.

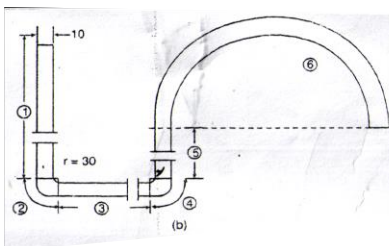


fig. 2.25

Inside radius of the bends = $30 - 10 = 20$ mm

$$\text{Length (1)} = 300 - (10 + 20) = 270 \text{ mm}$$

$$\text{Length (2)} = \frac{\alpha \cdot 2\pi}{360} (r + k)$$

Here $\alpha = 90^\circ$, now $\frac{r}{t} = 2$. let $K = 0.33 t$.

$$\text{Length (2)} = \frac{90}{360} \times 2\pi (20 + 0.33 \times 10) = 36.6 \text{ mm}$$

$$\text{Length (3)} = 200 - 2(10 + 20) = 140 \text{ mm}$$

$$\text{Length (4)} = \text{length (2)} = 36.6 \text{ mm}$$

$$\text{Length (5)} = 100 - (10 + 20) = 70 \text{ mm}$$

For length (6) inner radius = $150 - 10 = 140$ mm $\alpha = 50.25$ mm

Total developed length = (1) + (2) + (3) + (4) + (5) + (6)

$$= 270 + 36.6 + 140 + 36.6 + 70 + 45.25$$

$$= 100.45 \text{ mm}$$